

IT 320 Course Project
Semester-2, 1447H



Software Product Release <GreenRent>

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1 Chapter 1:

Agriculture has changed fast in recent years. This is mainly because farmers are using more technology to increase what they produce. Since heavy machinery is very expensive and capital-intensive, renting equipment has become a smart choice. It allows farmers to use the best tools without paying the full price to own them. This shift has made farming operations much more flexible and efficient [1].

However, even though this market is important, it is not running efficiently right now. Small farmers find it hard to get machinery because prices are high and the process isn't clear. On the other side, equipment owners have a problem: underutilization. Their expensive assets often sit idle, which is a waste of money and resources. The main reason for this mess is retail sector and the lack of a single digital platform to connect everyone [2].

To fix this, a centralized web-based application would be the ideal solution to organize the market. Such technological solutions are critical here because they act as a bridge. A digital platform helps close the information gap, makes prices transparent, and lowers costs. Locally, this alleviates the burdens on small farmers and supports Saudi Vision 2030 goals for food security. Globally, it puts the sharing economy into practice, reducing waste and making better use of assets [3].

This document outlines the proposal for GreenRent. It starts with The Problem (1.1) and our proposed Solution (1.2). Then, it covers The Product (1.3) details—vision, roadmap, and scope—followed by the Scrum Team (1.4). The next section will dig deeper into why this platform is so necessary.

1.1 The Problem

Agricultural equipment such as tractors, harvesters, and advanced irrigation systems plays a crucial role in improving productivity and efficiency in modern farming. However, these machines are often expensive and require substantial financial investment. Consequently, many small and medium-sized farmers face significant challenges in accessing such equipment, particularly when it is only needed for short or seasonal periods.

For instance, a small-scale farmer may require a harvester for only a few days during the harvest season. Purchasing such equipment is economically unjustifiable due to its high cost, while renting through traditional channels can be difficult because of limited availability, unclear pricing, or the absence of nearby rental providers. At the same time, many equipment owners possess machinery that remains unused for extended periods, resulting in inefficient utilization of valuable agricultural resources.

This situation highlights a clear mismatch between the demand for agricultural equipment and its actual availability. The lack of an organized and easily accessible approach to managing short-term equipment usage contributes to higher operational costs for farmers and inefficient utilization of existing assets. Addressing this gap represents a valuable opportunity to enhance resource utilization and promote more sustainable agricultural practices.

1.2 The Solution

This problem will be addressed by developing a digital platform called GreenRent that brings together equipment owners and farmers or individuals within one organized system.

The solution is based on creating a centralized marketplace where equipment owners can list their agricultural machinery with detailed information such as type, price, location, and availability period. At the same time, farmers will be able to search, filter, and request equipment based on their specific needs and rental duration.

The platform will provide essential services including availability tracking, transparent pricing display, booking management, and a review and rating system to ensure trust between both parties. By structuring the rental process digitally, the platform eliminates the difficulty of finding available equipment through traditional or informal channels.

It is expected that this platform will simplify the process of accessing agricultural equipment, reduce the time and effort spent searching for machinery, and lower operational costs for small and medium farmers. In addition, it will help equipment owners utilize idle machinery more efficiently, increasing overall resource optimization.

GreenRent specifically targets the creation of a reliable and accessible platform that connects renters and lessors, ensuring smoother transactions and improving efficiency in the agricultural sector.

1.3 The Product

1.3.1 Product Vision

For Farmers, small business owners, and individuals **Who** Need temporary access to agricultural equipment without the high cost of purchasing machines **The** GreenRent is a Web-based rental platform **That** Allows equipment owners to list their machines for rent and enables users to easily search, filter, and rent agricultural equipment for short-term use **Unlike** MachineryLink Sharing **Our Product** Provides a trusted platform with reviews, ratings, and smart filters (type, price, location) to make agricultural equipment rental affordable, reliable, and accessible.

1.3.2 Product Roadmap

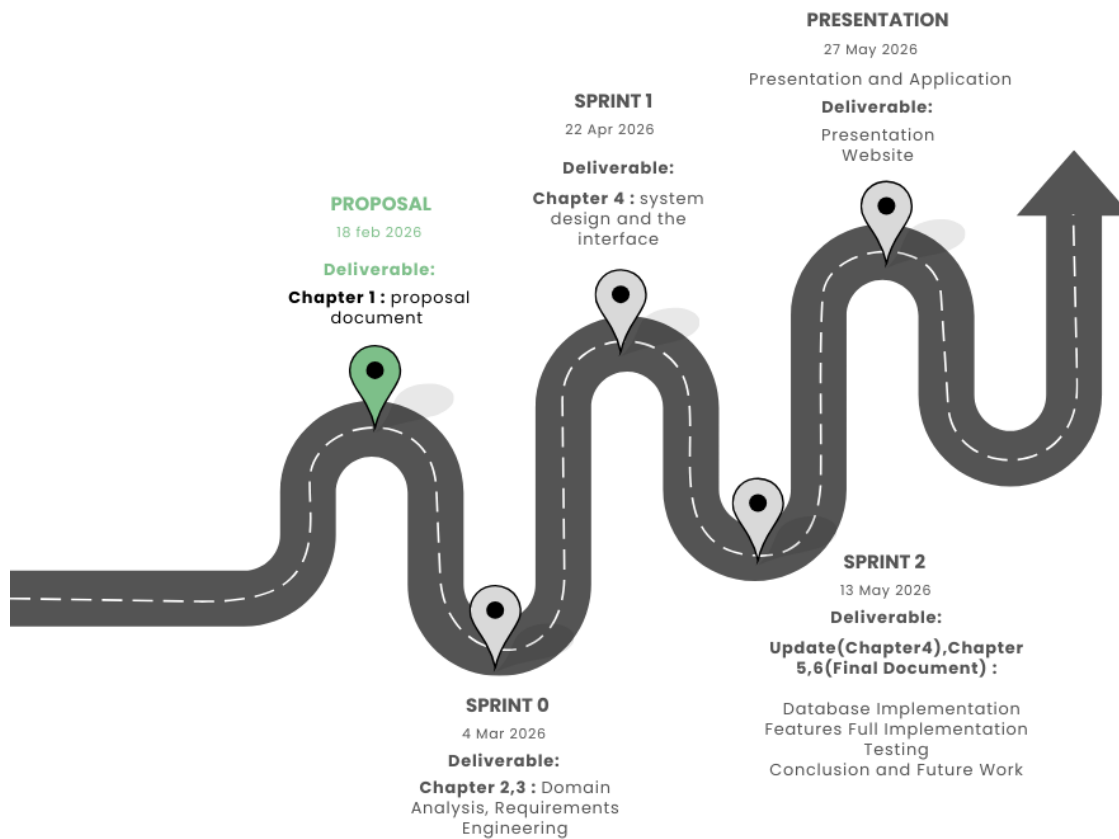


Figure 1 : RoadMap

1.3.3 Objectives

The objectives of the GreenRent project describe the goals that the team aims to achieve through the development of the system. These objectives are divided into product objectives, project objectives, and learning objectives in order to clearly define the value of the system, the development process, and the expected learning outcomes.

Product Objectives

The main objective of the GreenRent system is to provide a practical and easy-to-use platform that supports the process of renting agricultural equipment. The product objectives focus on enabling users to perform the essential tasks required to use the system efficiently. These objectives include:

- Allow users to register and create an account in the system in order to access available services.
- Enable registered users to log in to their accounts securely and log out when needed.
- Allow equipment owners to add agricultural equipment to the system by providing basic details such as name, type, price, location, and description.
- Enable equipment owners to edit the information of their listed equipment when updates are required.
- Allow equipment owners to delete their equipment listings from the system.
- Allow users to view detailed information about available agricultural equipment.
- Enable users to search for specific equipment easily.
- Allow users to filter available equipment based on criteria such as type, price, and location.
- Support a review and rating feature that allows users to share feedback after renting equipment, which helps improve trust among users.
- Support basic rental operations by allowing users to request equipment rentals and manage equipment availability.

Project Objectives

The project objectives focus on the steps required to successfully develop the GreenRent system. These objectives include:

- Understand the agricultural equipment rental domain and identify user needs.

- Define and document system requirements.
- Design simple and user friendly interfaces.
- Develop and implement the core functionalities of the system.
- Create a database that supports system operations.
- Test the system to ensure correctness and usability.
- Work collaboratively as a team throughout the project lifecycle.

Learning Objectives

The learning objectives describe the knowledge and skills that the team aims to gain during the project.

These objectives include:

- Gain practical experience in developing a web based software system.
- Apply software engineering concepts learned during the course.
- Improve teamwork, communication, and time management skills.
- Strengthen problem solving abilities by working on a real-world problem.
- Learn how to document and present a software project in a clear and structured manner.

1.3.4 Scope

GreenRent is a web-based application designed to facilitate the rental of agricultural equipment within the city of Riyadh . The website will be accessible through modern web browsers on desktop and mobile devices and will fully support Google Chrome, Safari, and Microsoft Edge. The system will support the English language only. GreenRent aims to serve both farmers who need agricultural equipment and equipment owners who wish to rent out their machinery.

The website will provide users with the ability to register, log in, and log out at any time, view detailed information, and also will allow equipment owners to add, edit, and delete agricultural equipment listings, including managing availability and basic equipment information. Renters will be able to

reserve equipment for specific dates and times and provide comments or reviews. The system will support basic editing and deletion of listings, reservations, and user feedback.

In addition, users will have the ability to search for available equipment, apply filters based on type, price, and location, and leave reviews and ratings for rented equipment and owners, also the GreenRent system will include an Admin role responsible for managing and monitoring the platform. The Admin will oversee user accounts, equipment listings, reservations, and reviews to ensure compliance with platform policies.

The Admin will have the authority to review ratings and user feedback and take action when necessary. If an equipment listing consistently receives low ratings or repeated complaints, the Admin may remove or deactivate it. Similarly, user accounts may be suspended in cases of misuse or multiple negative reports.

The Admin's role is limited to moderation and basic platform management and does not involve direct participation in rental transactions.

On the other hand, the GreenRent website will not include features that allow users to directly contact equipment owners. Furthermore, logistics related services such as equipment delivery, transportation, and physical handling are not included within the scope of this project. Any additional languages, or advanced operational features are considered outside the scope and will be addressed in future work and development.

1.4 The Scrum Team

Table 1: The Scrum Team

Scrum Team	
Product Owner:	Haya Alomar
Developers:	Sara Aljuraybah Dalal Alghumlas Deema Alquwaei Nora almousa
Scrum Master (SM):	I. Ghaida AL Fayeze
Stakeholders:	I. Ghaida Al Fayeze

2 Chapter 2: Domain Analysis

2.1 Literature Review

Competitor 1: Al Marwan Machinery [5]: A prominent heavy machinery rental company operating in Saudi Arabia, providing services primarily through a B2B model.

Strengths:

- Provides a professional local platform with a massive inventory of high-quality machinery.
- Offers integrated equipment delivery and logistics services to the customer's site.

Limitations:

- Does not provide a full checkout system with real-time pricing; users must submit a quote request to complete the rental.

- Does not support a Peer-to-Peer (C2C) model, preventing individual equipment owners from listing their machinery.

Competitor 2: DOZR [6]: An online marketplace for heavy and agricultural equipment rentals, primarily connecting contractors with commercial rental businesses.

Strengths:

- Provides a highly streamlined online booking process with transparent pricing.
- Offers advanced search and filtering features to find specific machinery quickly.

Limitations:

- Operates mainly on a B2B model, focusing on commercial rental houses rather than individual owners.
- Lacks an integrated user-to-user review and rating system to build trust between individuals.

Table 2 -1: Comparison between existing systems and GreenRent

Feature	Al Marwan	DOZR	GreenRent
Add and manage personal equipment listing			✓
Filter by price, type and location	✓	✓	✓
Submit and view user ratings and reviews			✓

Equipment delivery and logistics services	✓	✓	
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As shown in Table 2-1, GreenRent fills a significant market gap by adopting a Peer-to-Peer (C2C) model, unlike the primarily B2B approaches of Al Marwan and DOZR. While competitors focus on commercial rental businesses, GreenRent empowers individual equipment owners to list and manage their machinery directly. Moreover, it enhances trust through an integrated user-to-user review system and provides a fully digital, transparent booking process with instant interaction instead of manual quote requests. This positioning enables GreenRent to create a more accessible and community-oriented rental platform in the Saudi market.

3 Chapter 3: Requirements Engineering

3.1 Requirements Elicitation and Analysis

As part of the requirements engineering process for the GreenRent platform, we began by eliciting and documenting the system requirements to ensure a clear understanding of stakeholders' needs and expectations. Although various elicitation techniques exist, we selected **interviews** and **questionnaires** as the most appropriate methods for directly engaging with our target users.

We conducted five interviews (See Appendix A: Interviews) with key stakeholders, including farmers and equipment owners, to collect qualitative insights about rental practices, platform expectations, and desired system features. In addition, a structured questionnaire (See Appendix B: Questionnaire) was distributed to gather quantitative data from a broader group of potential users.

These elicitation methods provided a solid foundation for defining and structuring the system requirements.

Based on the five conducted interviews (see Appendix A), several recurring themes emerged regarding agricultural equipment rental through an online platform.

First, the majority of participants (Interviews 1, 2, and 4; see Appendix A) do not currently rent out their equipment formally. While Interviews 3 and 5 indicated occasional informal rentals based on verbal agreements and personal trust, none of the participants reported using official contracts. This suggests that the current rental practices lack structure and legal protection.

A consistent concern across all five interviews (Q2; see Appendix A) was the risk of equipment damage and late returns. Participants repeatedly emphasized the financial impact of machinery breakdowns and the disruption caused by delayed returns. Additionally, concerns about delayed or incomplete payments were mentioned in Interviews 1, 3, and 5. These findings indicate that operational and financial risks are the primary barriers preventing wider participation in equipment rental.

Trust and security were identified as critical decision factors (Q3; see Appendix A). All participants stressed the importance of secure payment systems, verified renters, and clear rental agreements. Interviews 3 and 4 specifically highlighted the need for electronic contracts and protection mechanisms against damage. This demonstrates a strong demand for legal and financial safeguards within the platform.

Regarding payment preferences (Q6; see Appendix A), all participants preferred documented and secure payment methods such as online platform payments or bank transfers. None of the interviewees preferred cash transactions. This reflects a clear need for traceable and secure financial processing.

The rating system was unanimously considered important (Q5; see Appendix A). All participants agreed that a review system increases trust, allows assessment of renter credibility, and improves accountability. This indicates that transparency between users is essential for platform adoption.

Participants also emphasized the importance of displaying clear and detailed equipment information (Q4; see Appendix A), including equipment type, condition, price, availability schedule, and location. Several interviewees also stressed the need to clarify whether the equipment includes an operator.

Finally, technical and platform-related concerns were mentioned in Interviews 1, 4, and 5 (Q7; see Appendix A), including fear of fraud, unclear terms, high platform fees, technical issues, and slow customer support. Ease of use was also highlighted as an influencing factor for adoption.

Overall, the interviews suggest that while agricultural equipment owners are open to using an online rental platform, adoption depends heavily on trust, financial security, protection mechanisms, and transparency (see Appendix A).

The questionnaire included ten main questions covering user roles, rental experience, difficulty in finding equipment, decision factors, importance of reviews, platform adoption, concerns, usage frequency, trust factors, and suggested improvements. According to Graph 1, 70% of respondents are farmers, 20% are equipment owners, and 10% are both, indicating that farmers represent the primary target users of the system. Graph 2 shows that 70% have previously rented agricultural equipment, confirming an existing and active demand for rental services. However, Graph 3 reveals that 50% of participants find it difficult and 20% find it very difficult to locate equipment for rent, meaning that 70% experience significant difficulty, which highlights a clear market gap. In Graph 4, price and availability are identified as the most important factors (30% each), followed by equipment condition (20%), emphasizing the need for transparent pricing and accurate availability management within the system. Furthermore, Graph 5 indicates that 40% consider reviews very important and 50% consider them important, demonstrating that 90% value ratings as a trust mechanism. Graph 6 shows that 70% would use a platform like GreenRent and 30% might use it, reflecting strong potential adoption. Nevertheless, Graph 7 identifies payment security (40%) and lack of trust (30%) as the main concerns, followed by equipment damage and fraud (20% each), highlighting the necessity of strong security measures. According to Graph 8, most users would use the platform seasonally (50%) or monthly (30%), suggesting demand peaks during agricultural seasons. Graph 9 demonstrates that clear rental agreements (40%), secure payment systems (30%), and verified accounts (20%) are the most important trust-building features. Finally, Graph 10 suggests additional improvements such as direct communication between users, availability calendars, insurance options, delivery services, and customer support, which can further enhance system usability and reliability. Overall, the findings confirm the need for a secure, transparent, and user-centered agricultural equipment rental platform that prioritizes trust, pricing clarity, availability, and transaction security to ensure successful adoption.

To further understand the perspective of equipment owners and complete the market analysis, a second questionnaire was specifically distributed to 10 agricultural equipment owners. According to Graph 11, half of the participants (50%) indicated that they currently rent out their equipment but informally, while 30% rent it out regularly, and 20% have never rented it out. This confirms that current market practices lack organization and rely heavily on personal relationships and verbal agreements.

Regarding challenges and concerns, Graph 12 illustrates that the biggest fear preventing owners from renting out their machinery is equipment damage or misuse (50%), followed by the renter returning the equipment later than the agreed time (20%), and a lack of trust in unknown renters (20%). These figures clearly highlight the scale of operational risks perceived by the owners. To address these concerns, Graph 13 reviews the most requested technical features; the need for a secure online payment system tied with the ease of managing equipment availability (calendar) at 30% each, followed by electronic contracts and rating systems at 20% each.

When asked about options that enhance their sense of security, Graph 14 showed that 40% of participants consider "requiring verified renter identities" as the most important factor, while 30% preferred a comprehensive security package that includes verification, an advance deposit, and insurance. Regarding financial transactions, Graph 15 indicates that the majority (60%) prefer receiving rental profits through a secure online platform, compared to 30% who prefer direct bank transfers, which emphasizes the necessity of integrating a reliable payment gateway within the system.

Finally, regarding equipment availability times, Graph 16 demonstrates that 50% of owners expect to make their equipment available seasonally, which perfectly matches the peak demand periods of renters (as shown in the first questionnaire), while 20% intend to make it available year-round. Furthermore, open-ended responses suggested additional features to enhance platform efficiency, such as automated SMS reminders to prevent late returns, a direct chat feature between both parties, and detailed maintenance logs to track the equipment's condition.

Based on the comprehensive analysis of the interviews and both questionnaires, several core system requirements for the GreenRent platform have been definitively established. First, there is a clear market demand driven by the difficulty renters face in finding equipment (70%) and the informal,

unstructured nature of current renting practices among equipment owners. Second, trust, security, and risk mitigation are the most critical factors for platform adoption. Both renters and owners share strong mutual concerns regarding payment security, equipment damage, fraud, and late returns. Therefore, the system must implement strict security measures, including a secure online payment gateway, verified user accounts, electronic rental agreements, and a transparent rating and review system (highly valued by 90% of renters). Third, because equipment usage and availability are predominantly seasonal for both parties, the platform must provide an efficient availability calendar and robust search filters focusing on price, condition, and location. By integrating these specific functional and non-functional requirements, GreenRent will successfully address user concerns and establish a reliable, secure, and user-centered Peer-to-Peer (C2C) agricultural equipment rental marketplace.

نسخ الرسم البياني

?What is your role

10 ردود

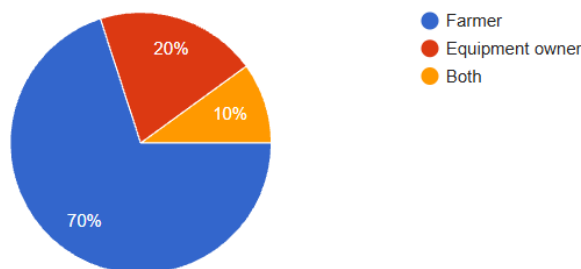


Figure 2 : **What is your role?** Identifies participants' professional background and role.

Graph 1

نسخ الرسم البياني

?Have you ever rented agricultural equipment

10 ردود

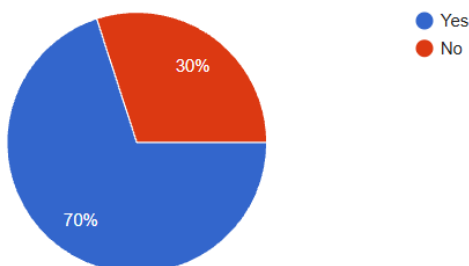


Figure 3 : **Have you ever rented agricultural equipment?** Assesses prior experience with equipment rental.

Graph 2

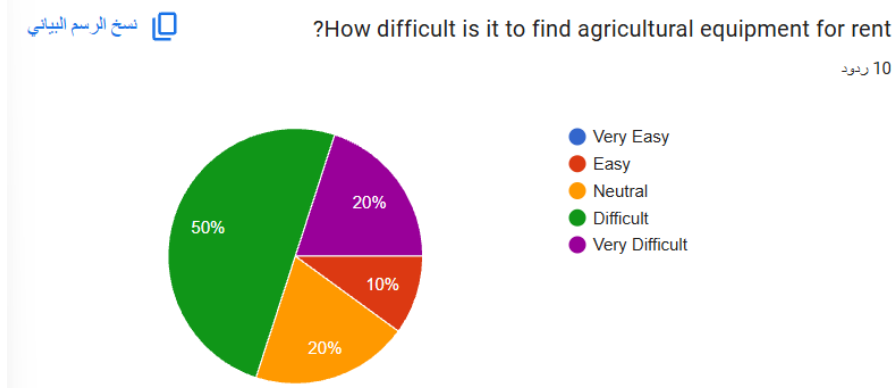


Figure 4 :How difficult is it to find agricultural equipment for rent? Explores the difficulty of finding rental equipment.

Graph 3

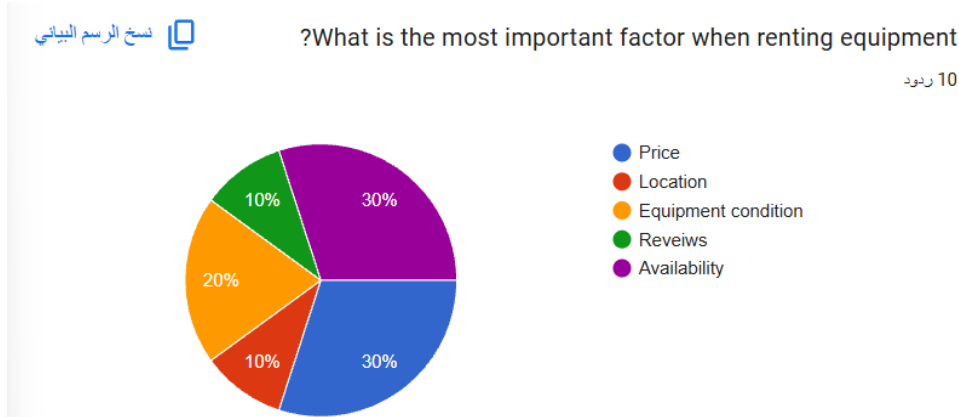


Figure 5: What is the most important factor when renting equipment? Identifies the key priorities influencing rental decisions.

Graph 4

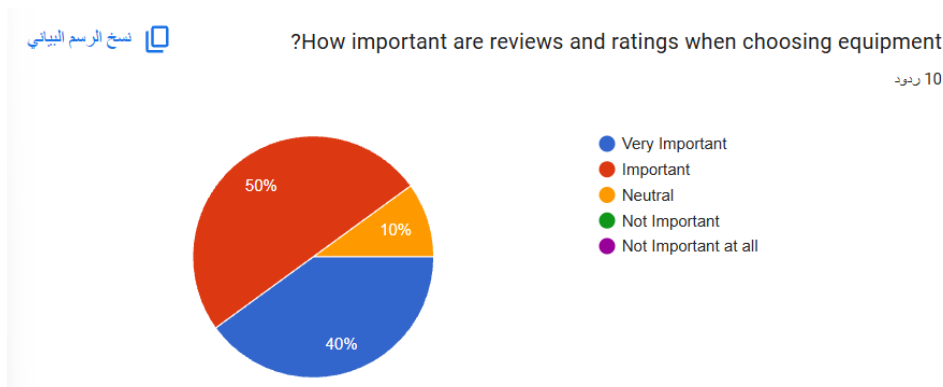


Figure 6 : How important are reviews and ratings when choosing equipment? Measures the value users place on peer feedback.

Graph 5

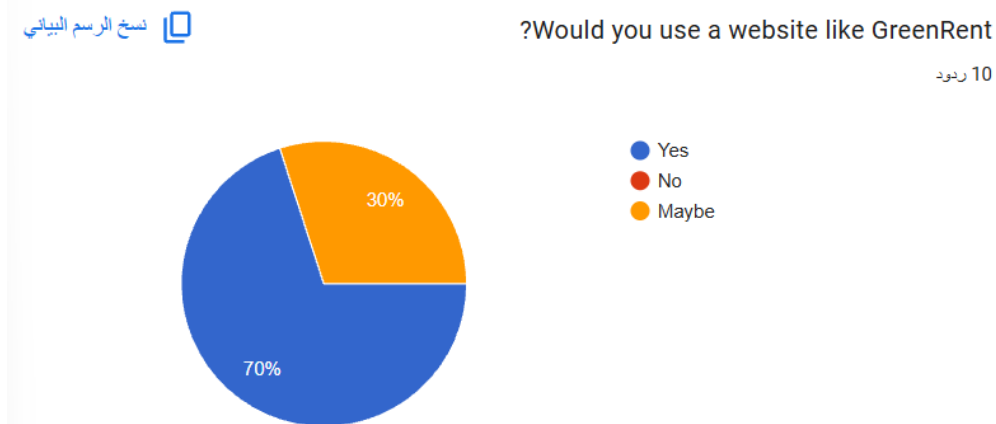


Figure 7 : What concerns do you have about online rental platforms? Identifies main barriers to using online rental services.

Graph 6

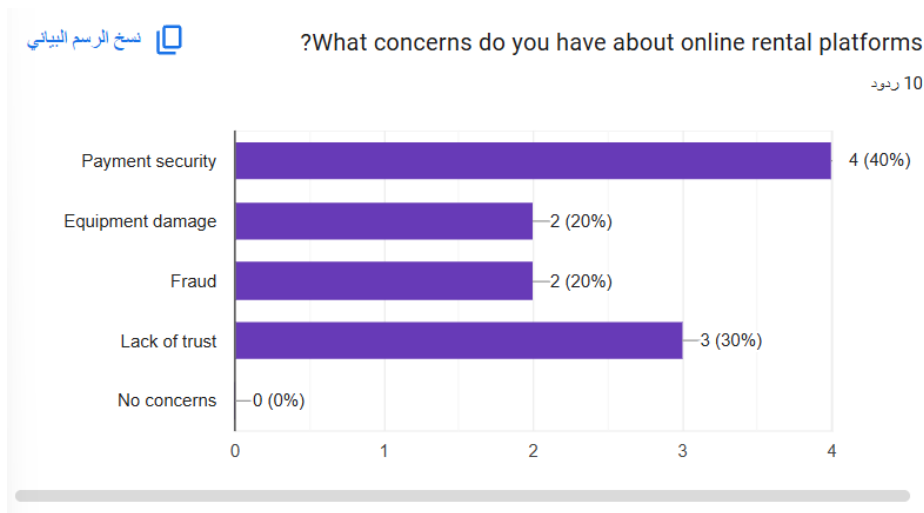


Figure 8 :How often would you use such a platform? Determines expected frequency of platform usage.

Graph 7

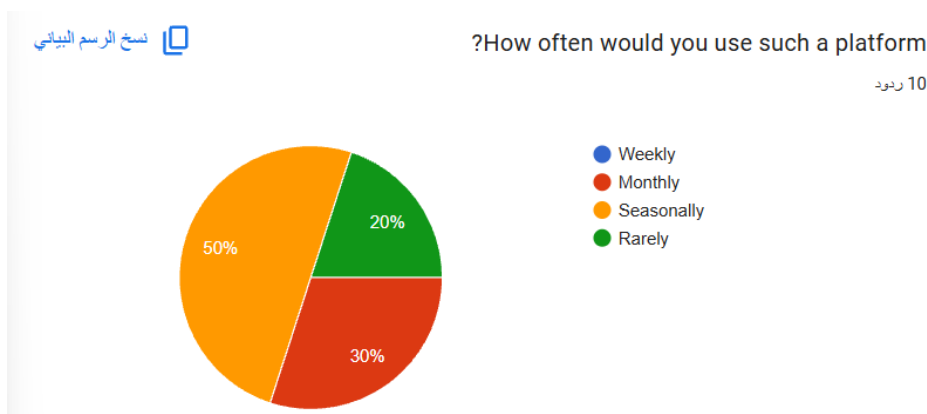


Figure 9 :What would make you trust an online agricultural rental platform? Uncovers key factors that build user trust.

Graph 8

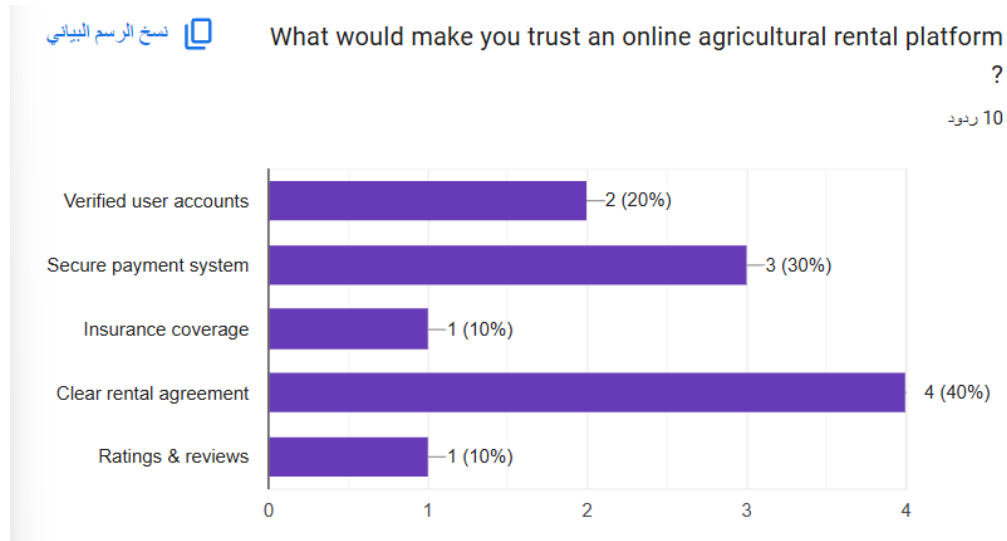


Figure 10 : What improvements or additional features would you suggest? Gathers ideas for future platform improvements.

Graph 9

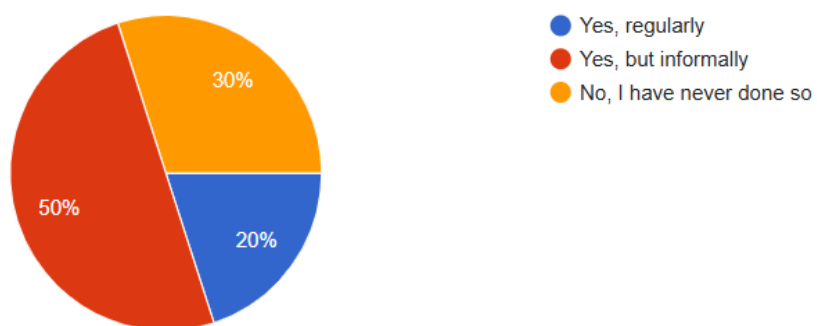


Figure 11: Participants' Previous Experience in Renting Out Agricultural Equipment

Graph 11

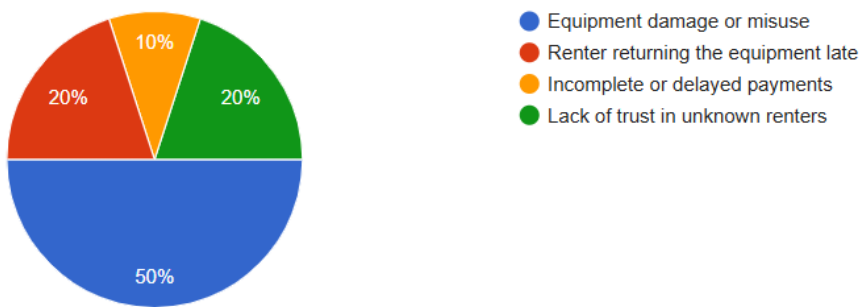


Figure 12: Main Challenges and Concerns Preventing Owners from Renting Out Equipment

Graph 12

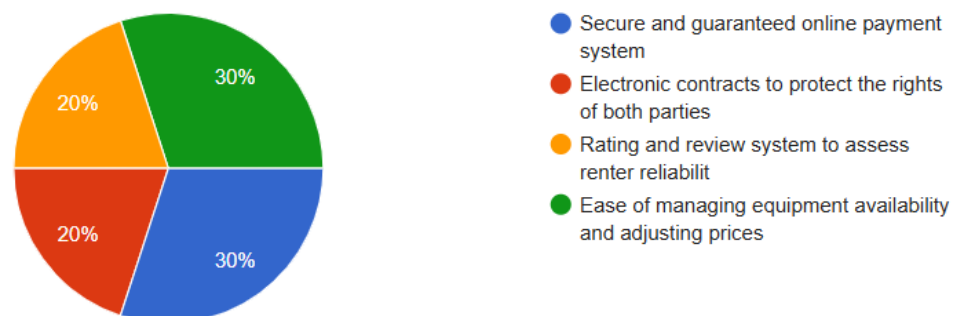


Figure 13 :Most Important Technical Features Desired by Equipment Owners

Graph 13

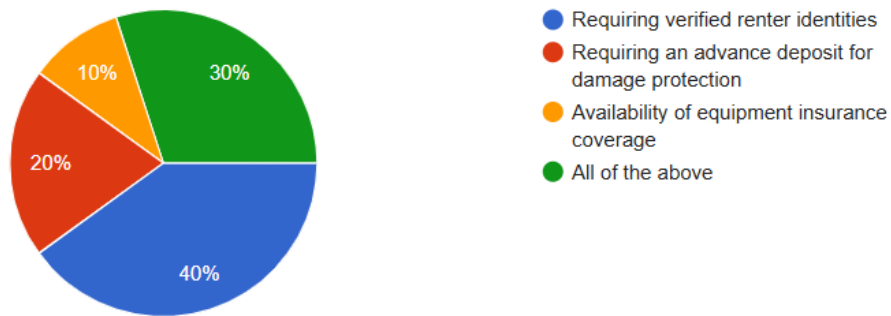


Figure 14 :Preferred Safety Measures for Listing Equipment on the Platform

Graph 14

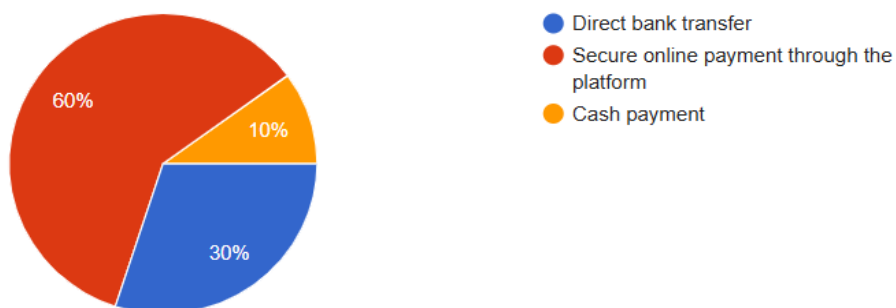
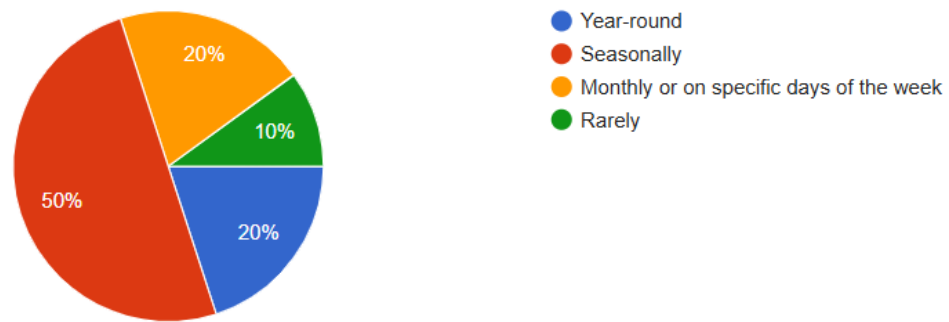


Figure 15 :Preferred Methods for Receiving Rental Profits

Graph :



Graph 16

Figure 16 :Expected Frequency of Equipment Availability for Rent

• Participants information:

Table 3: participants information

PARTICIPANT#	NAME	AGE	GENDER	NATIONALITY	LOCATION	EXPERIENCE IN FARMING
P1	Ali	65	Male	Saudi	sudair	15 years
P2	Mohammed	58	Male	Saudi	Huraymila	10 Years
P3	Abdulaziz	45	Male	Saudi	Al-Kharj	8 Years
P4	Ahmad	46	Male	Saudi	Riyadh	13 years
P5	Saleh	68	Male	Saudi	Al-Kharj	20 years

3.2 System Users

The primary users of the GreenRent system are farmers and agricultural equipment owners in Saudi Arabia, particularly in agricultural areas such as Najd and surrounding cities. Based on the collected data, most users are farmers (70%), while a smaller percentage are equipment owners (20%) and some are both (10%).

The target users are typically between 45 and 70 years old and have strong experience in farming, ranging from 8 to 20 years. Most of them are Saudi nationals and have practical agricultural

knowledge. However, their technical expertise varies, and many prefer simple and easy-to-use digital systems.

Farmers use the platform to search for available equipment at reasonable prices, especially during seasonal farming periods. They are mainly concerned about equipment availability, fair pricing, and secure payment methods. Equipment owners use the system to rent out their machinery safely and are concerned about equipment damage, late returns, and payment security.

Since many users are middle-aged or older, the system must be clear, simple, and trustworthy. Features such as secure payment, verified accounts, clear rental agreements, and ratings are essential to build trust and encourage adoption of the platform.

3.3 Use Case Diagram

This section presents the use case diagram of the GreenRent system and shows how different users interact with the system.

The system includes three main actors: Farmer, Equipment Owner, and Admin.

The Farmer can register, log in, search for equipment, and filter the results to find suitable equipment. The farmer can also view equipment details before making a reservation. In addition, the farmer can add a reservation, edit or cancel reservations, view previous reservations, and leave ratings and reviews after using the equipment. The farmer can also log out of the system.

The Equipment Owner is responsible for managing equipment listings. This includes adding new equipment, editing equipment information, deleting equipment, setting equipment availability, and managing pricing.

The Admin is responsible for managing the platform. The admin can manage user accounts, suspend users when necessary, manage reservations, moderate reviews and ratings, and moderate equipment listings such as deactivating listings if needed.

Overall, the use case diagram helps illustrate the main functionalities of the GreenRent system and the interactions between the users and the platform.

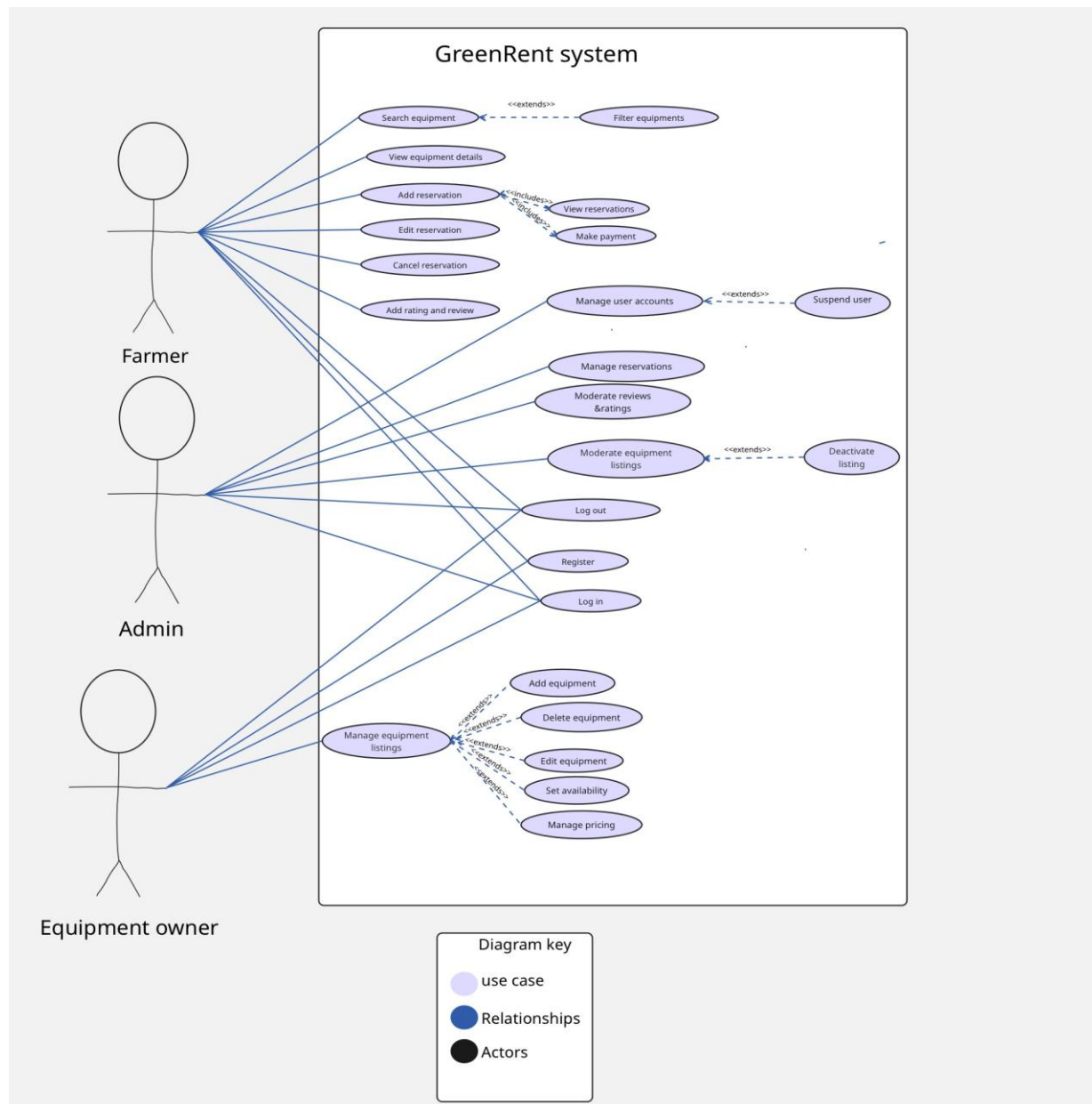


Figure 17 : Use case diagram

3.4 Product Backlog

Table 4: Product Backlog of GreenRent

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
As a new user, I want to register an account so that I can access rental services.	3	Feature	- The system displays a registration form with required fields (name, email, password, role). - The system validates that all required fields are completed. - If the email is already registered, the system shows an appropriate error message. - After successful registration, the user account is created. - The user is redirected to the login page (or dashboard). - The system stores the user information securely in the database.
As a registered user, I want to log in using my username and password so that I can securely access my account and personal information	2	Feature	- The system displays a login form with username and password fields. - The system verifies the entered credentials against stored user data. - If the username and password are correct, the user is redirected to their dashboard. - If the credentials are incorrect, an appropriate error message is displayed. - The user session is created after successful login.

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
			- The user cannot access protected pages without logging in. - The login process should be completed within 5 seconds.
As a registered user, I want to log out securely so that I can protect my account from unauthorized access.	2	Feature	- The system provides a visible "Log out" option for logged-in users. - When the user clicks "Log out", the session is terminated immediately. - After logging out, the user is redirected to the login (or home) page. - The user cannot access protected pages after logging out. - If the user tries to access a protected page after logout, the system redirects them to the login page. - The logout process completes successfully without errors.
As an equipment owner, I want to add my equipment details so that renters can view and rent it.	5	Feature	The system provides a form where the owner can enter equipment name, description, price, location, and availability.

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
			- The system validates that all required fields are completed before submission. - After successful submission, the equipment is stored in the database. - The equipment becomes visible to renters in the listing page.
As an equipment owner, I want to edit my equipment details so that I can update information when needed.	3	Feature	- The system allows the owner to access and modify existing equipment details. - The system validates required fields before saving updates. - After successful update, the modified information is stored in the database. - The updated equipment details are immediately visible to renters.
As an equipment owner, I want to delete my equipment so that it is no longer visible.	2	Feature	- The system allows the owner to select and delete a listed equipment item. - The system displays a confirmation message before deleting the equipment. - After confirmation, the equipment is permanently removed from the database.

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
			The deleted equipment is no longer visible to renters on the platform.
As a renter, I want to search equipment by name so that I can quickly find what I need.	3	Feature	- The system provides a search bar where the renter can enter equipment name keywords. - When a valid keyword is entered, matching equipment results are displayed. - If no equipment matches the keyword, the system displays a "No results found" message. - Search results are updated immediately after the renter submits the search query.
As a renter, I want to view detailed equipment information so that I can make a decision.	3	Feature	- The system displays detailed equipment information, including name, description, price, location, and availability. - The system displays the equipment condition and specifications clearly. - The system displays images of the equipment. - The equipment details page loads correctly when the renter selects an item.

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
As a renter, I want to filter equipment by type, price, and location so that I can narrow my choices.	3	Feature	- The system provides filter options for equipment type, price range, and location. - When the renter selects filter criteria, the displayed results update accordingly. - Multiple filters can be applied at the same time. - If no equipment matches the selected filters, a “No results found” message is displayed.
As a renter, I want to reserve equipment for specific dates so that I can use it during my required period.	5	Feature	- The system allows the renter to select available start and end dates for the equipment. - The system prevents booking if the selected dates are already reserved. - After successful reservation, a confirmation message is displayed. - The reservation details are saved and visible to both the renter and the equipment owner.
As an equipment owner, I want to receive secure online payment so	8	Feature	The system allows equipment owners to enable online payment for their listings.

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
that I can ensure guaranteed transactions.			The owner receives a confirmation notification after successful payment. The transaction details are recorded in the system. Payment is transferred to the owner's registered account.
As a renter, I want to leave a rating and review after renting so that I can share my experience.	2	Feature	- The system allows the renter to submit a rating (e.g., 1–5 stars) after completing a rental. - The renter can write and submit a written review. - Only renters who have completed a reservation can submit a rating and review. - The submitted rating and review are displayed on the equipment details page.
As an Admin, I want to view and manage user accounts so that I can ensure proper use of the platform.	3	Feature	- The system allows the Admin to view a list of all registered users. - The system displays user details (name, email, role, account status). - The Admin can change the account status (active / inactive).

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
			- Any changes made by the Admin are saved in the database. - The updated account status is immediately reflected in the system.
As an Admin, I want to suspend or deactivate user accounts when policy violations occur so that I can maintain platform integrity.	2	Feature	- The Admin can select a user account to suspend or deactivate. - The system requires confirmation before suspending the account. - After confirmation, the user account status changes to "Suspended" or "Inactive". - Suspended users cannot log in to the system. - The suspension action is recorded in the system.
As an Admin, I want to review equipment listings to ensure they comply with platform policies.	3	Feature	- The system allows the Admin to view all equipment listings. - The Admin can access equipment details (owner, price,

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
			description, location, availability). - The system displays listing status (active / inactive). - The Admin can identify listings that violate platform policies.
As an Admin, I want to remove or deactivate equipment listings that receive repeated complaints or low ratings so that I can maintain service quality.	3	Feature	- The Admin can view ratings and complaints related to each equipment listing. - The system highlights listings with consistently low ratings. - The Admin can deactivate or remove a listing. - The system requires confirmation before removal. - Removed listings are no longer visible to users. - The action is recorded in the system.

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria: The conditions of satisfaction that must be met for that item to be accepted.
As an Admin, I want to monitor reviews and ratings to identify misuse or policy violations.	2	Feature	- The system allows the Admin to view all submitted reviews and ratings. - The Admin can identify inappropriate or abusive content. - The Admin can take appropriate action if needed
As a renter, I want search results to load within 3 seconds so that I can quickly find equipment.	3	Technical work	- Search results load within 3 seconds under normal load. - System performance is tested before deployment.
As a user, I want the platform to be available at least 99% of the time so that I can access rental services whenever I need them.	3	Technical work	- The system uptime must be at least 99% per month. - Scheduled maintenance must be announced in advance. - In case of downtime, users receive a notification message. - The system automatically logs downtime incidents.

4 Chapter 4: System Design and Development

4.1 System Architecture

The system architecture of GreenRent plays an essential role in determining how the platform delivers its functionality, manages data, and supports user interaction. Since GreenRent is a web-based rental platform that connects equipment owners, renters, and administrators in one centralized system, the selected architecture must support accessibility, maintainability, scalability, and clear separation of responsibilities. GreenRent allows users to register, log in, browse equipment, search and filter listings, view equipment details, make reservations, manage listings, and submit reviews and ratings. These functions require continuous interaction between the user interface, the application logic, and the system database.

After analyzing the nature of the system and reviewing suitable architectural patterns, the GreenRent team selected the **Client–Server Architecture** as the most appropriate architecture for the platform. In the Client–Server pattern, the system is organized into three main parts: the client, the server, and the network. The client requests services, the server processes these requests and provides the

required services, and the network enables communication between both sides. This architectural style is widely used for online systems and web applications.

In GreenRent, the **client side** represents the web interface that is accessed by the main users of the system, including renters, equipment owners, and admins. Through the client side, users can interact with the platform by viewing the home page, browsing available equipment, adding or editing equipment listings, making reservations, and managing their accounts. The client side is responsible for presenting information in a clear and user-friendly manner, collecting user input, and sending requests to the server for processing.

The **server side** is responsible for handling the core functionality of the system. It contains the application logic that manages registration, login, equipment listing, search and filtering operations, reservation processing, review and rating management, and admin moderation functions. It also communicates with the database to store, retrieve, update, and delete system data. In GreenRent, the server acts as the central processing unit of the application and ensures that user requests are handled correctly and that the platform data remains consistent and up to date.

The **database** stores the main system entities required for the operation of GreenRent. According to the system data design, the platform manages data related to users, equipment, reservations, reviews, and payments. These data entities are essential for supporting the platform's main services, such as listing agricultural equipment, checking availability, confirming reservations, recording reviews, and managing secure rental transactions.

The GreenRent team chose the Client–Server Architecture for several reasons. First, it provides a clear separation between the presentation layer and the business logic and data management parts of the system, which improves maintainability and makes future updates easier. Second, it is suitable for a web-based platform where multiple users may access the system from different locations through browsers and devices. Third, it supports scalability, allowing the platform to grow as the number of users, equipment listings, and rental requests increases. Finally, centralizing the business logic and data storage on the server side improves control, consistency, and security, which is especially important for sensitive operations such as user authentication, reservation management, and payment processing. These design considerations are consistent with the role of architectural

design in defining the main system components and their relationships in a way that satisfies both functional and non-functional requirements.

Other architectural patterns were considered but were not selected. The **MVC architecture** is useful when the same data must be presented in different ways, but for GreenRent, the project focus is on overall system organization rather than dividing the application internally into model, view, and controller components. The **Layered Architecture** is also a strong option because it organizes the system into layers, but for this project, the Client–Server pattern provides a simpler and more direct representation of how the web platform operates between users and the centralized backend service. The **Repository Architecture** was not selected because GreenRent is not primarily a repository-centered engineering environment where independent tools communicate only through shared data. Similarly, the **Pipe and Filter Architecture** was not chosen because it is more suitable for data-processing systems and is not appropriate for interactive online platforms like GreenRent.

In conclusion, the Client–Server Architecture was selected as the best architectural solution for GreenRent because it matches the nature of the platform as an interactive web-based rental system. It provides an effective structure for communication between users and the system, supports centralized data management, and offers a practical foundation for implementing the main services of GreenRent in a secure, organized, and scalable way.

4.2 Class Diagram

The following Class Diagram illustrates the static structure of the GreenRent system. It highlights the main classes, their attributes, operations, and the relationships between them.

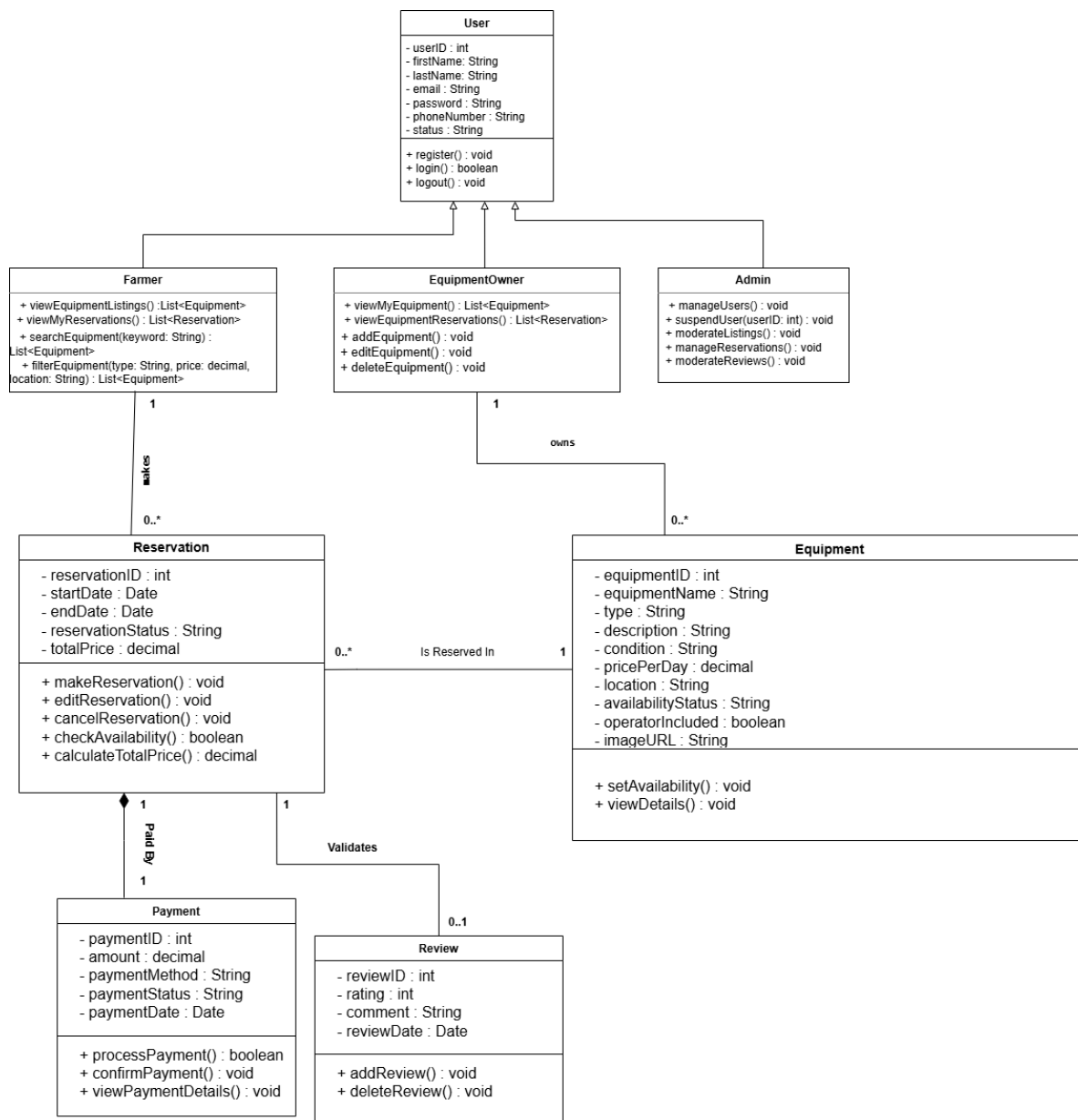


Figure 18 Class Diagram

4.3 Data Design

4.3.1 Entity Relationship Diagram (ERD)

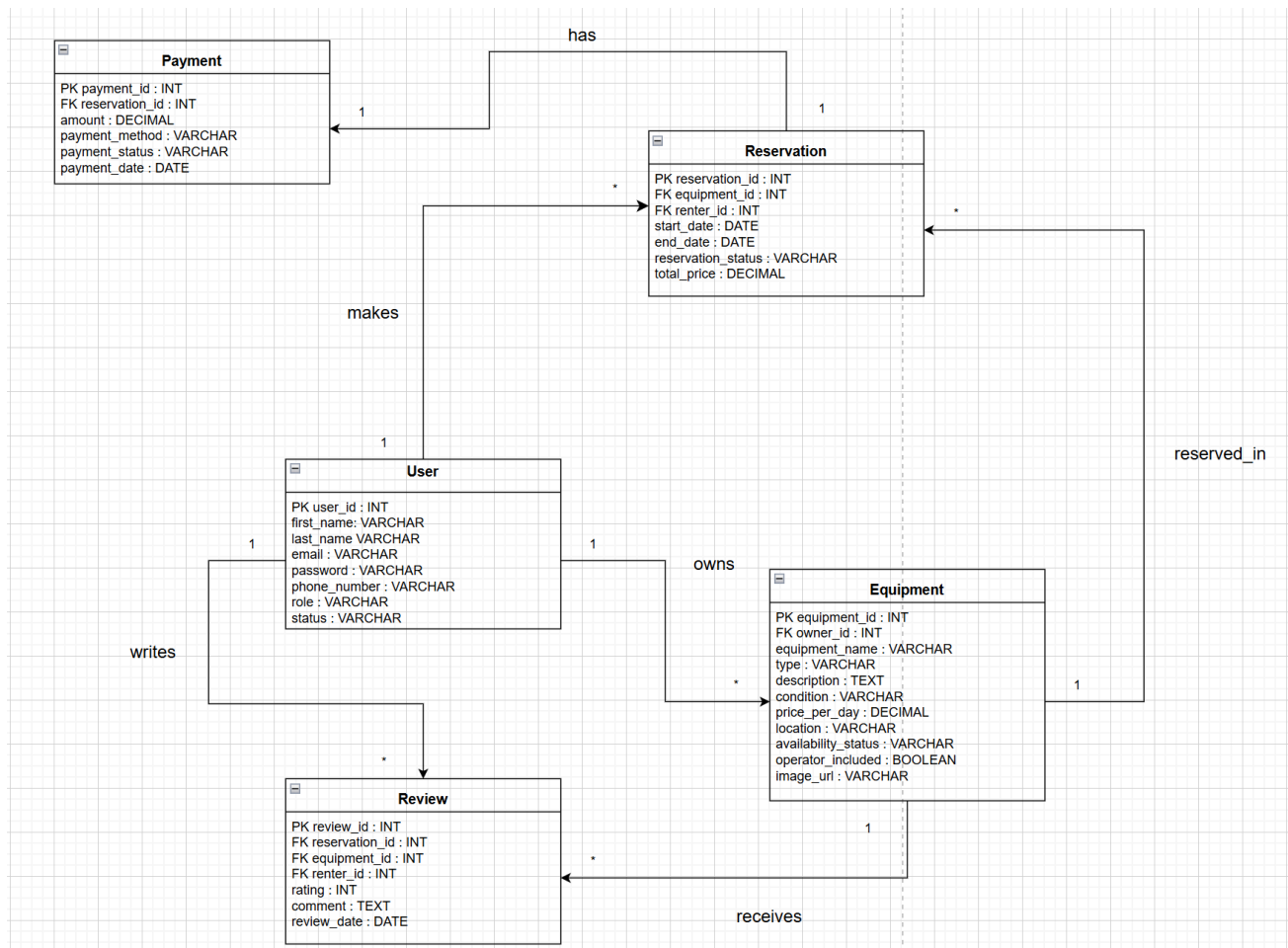


Figure 19: Entity Relationship Diagram (ERD) for GreenRent System

4.3.2 Relational Schema

User

```

User(
    user_id PK,
    full_name,
    email,
    password,
    phone_number,

```

```
role,
status
)
```

Equipment

```
Equipment(
    equipment_id PK,
    owner_id FK → User.user_id,
    equipment_name,
    type,
    description,
    condition,
    price_per_day,
    location,
    availability_status,
    operator_included,
    image_url
)
```

Reservation

```
Reservation(
    reservation_id PK,
    equipment_id FK → Equipment.equipment_id,
    renter_id FK → User.user_id,
    start_date,
    end_date,
    reservation_status,
    total_price
)
```

Review

```
Review(
    review_id PK,
    reservation_id FK → Reservation.reservation_id,
    equipment_id FK → Equipment.equipment_id,
    renter_id FK → User.user_id,
    rating,
    comment,
    review_date
)
```

Payment

```
Payment(
    payment_id PK,
    reservation_id FK → Reservation.reservation_id,
    amount,
    payment_method,
)
```

```
payment_status,  
payment_date  
)
```

4.3.3 Data Dictionary

The following data dictionary describes the main database tables used in the GreenRent system. It includes the table attributes, data types, descriptions, and key constraints. This dictionary helps clarify how system data is structured and supports the core functions of the platform, including equipment listing, searching, reservation, payment, and review management.

Data Dictionary for User

Table 5: User Data Dictionary

Attribute	Data Type	Description	Key	Required
user_id	INT	Unique identifier for each user	PK	Yes
full_name	VARCHAR	Full name of the user	-	Yes
email	VARCHAR	User email address	-	Yes
password	VARCHAR	User password	-	Yes
phone_number	VARCHAR	User phone number	-	Yes
role	VARCHAR	User role in the system (renter, owner, admin)	-	Yes
status	VARCHAR	Current account status (active, inactive, suspended)	-	Yes

Data Dictionary for Equipment

Table 6: Equipment Data Dictionary

Attribute	Data Type	Description	Key	Required
equipment_id	INT	Unique identifier for each equipment item	PK	Yes
owner_id	INT	Identifier of the equipment owner	FK	Yes
equipment_name	VARCHAR	Name of the equipment	-	Yes
type	VARCHAR	Type or category of equipment	-	Yes
description	TEXT	Detailed description of the equipment	-	Yes
condition	VARCHAR	Current condition of the equipment	-	Yes
price_per_day	DECIMAL	Rental price per day	-	Yes
location	VARCHAR	Location where the equipment is available	-	Yes
availability_status	VARCHAR	Availability status of the equipment	-	Yes
operator_included	BOOLEAN	Indicates whether an operator is included	-	No
image_url	VARCHAR	Path or URL of the equipment image	-	No

Data Dictionary for Reservation

Table 7: Reservation Data Dictionary

Attribute	Data Type	Description	Key	Required
-----------	-----------	-------------	-----	----------

reservation_id	INT	Unique identifier for each reservation	PK	Yes
equipment_id	INT	Identifier of the reserved equipment	FK	Yes
renter_id	INT	Identifier of the renter who made the reservation	FK	Yes
start_date	DATE	Reservation start date	-	Yes
end_date	DATE	Reservation end date	-	Yes
reservation_status	VARCHAR	Status of the reservation (pending, confirmed, cancelled, completed)	-	Yes
total_price	DECIMAL	Total reservation cost	-	Yes

Data Dictionary for Review

Table 8: Review Data Dictionary

Attribute	Data Type	Description	Key	Required
review_id	INT	Unique identifier for each review	PK	Yes
reservation_id	INT	Identifier of the related reservation	FK	Yes
equipment_id	INT	Identifier of the reviewed equipment	FK	Yes
renter_id	INT	Identifier of the renter who submitted the review	FK	Yes
rating	DECIMAL	Numeric rating given by the renter	-	Yes
comment	TEXT	Written review comment	-	No
review_date	DATE	Date when the review was submitted	-	Yes

Data Dictionary for Payment

Table 9: Payment Data Dictionary

Attribute	Data Type	Description	Key	Required
payment_id	INT	Unique identifier for each payment transaction	PK	Yes
reservation_id	INT	Identifier of the related reservation	FK	Yes
amount	DECIMAL	Amount paid for the reservation	-	Yes
payment_method	VARCHAR	Payment method used (online payment, bank transfer)	-	Yes
payment_status	VARCHAR	Current payment status (paid, pending, failed, refunded)	-	Yes
payment_date	DATE	Date of the payment transaction	-	Yes

4.4 Component Design

This section presents the design of three key components in the GreenRent system. The selected components are Add reservation , Filter equipments, and Add equipment. Each component is illustrated using a flowchart and supported with pseudocode

Add reservation

Flowchart :

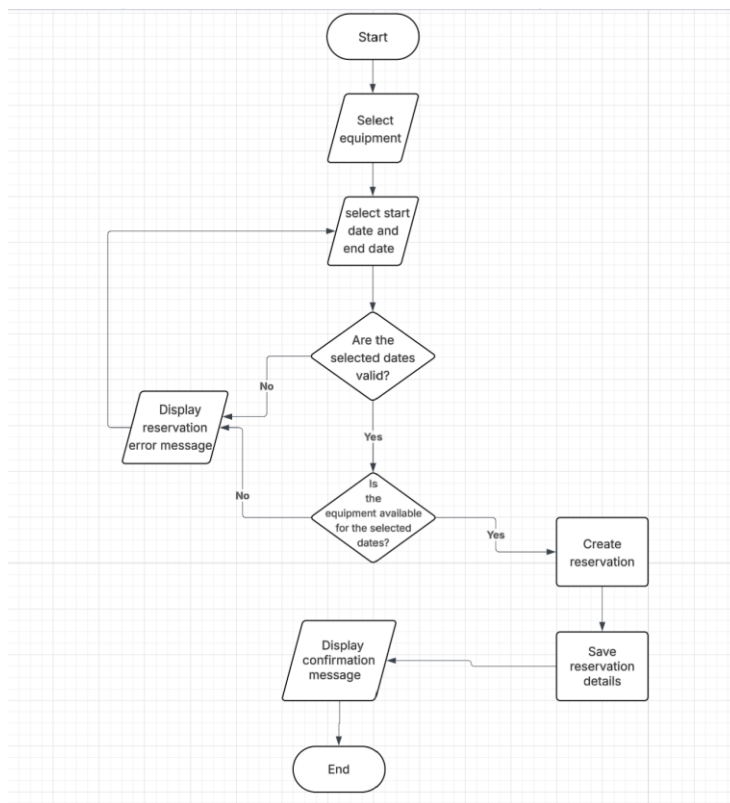


Figure 20 :Flowchart for Reserve Equipment

Pseudocode:

BEGIN ReservationProcess(renterID, equipmentID)

isConfirmed = False

REPEAT

INPUT startDate, endDate

IF (startDate < Today OR endDate <= startDate) THEN
DISPLAY "Reservation error message"

ELSE IF (EquipmentIsBusy(equipmentID, startDate, endDate)) THEN
DISPLAY "Reservation error message"

ELSE

CREATE new reservation record

SET reservationStatus = "Confirmed"

SAVE reservation details to Database

DISPLAY "Success: Your reservation has been confirmed!"

isConfirmed = True

END IF

UNTIL isConfirmed = True

END

Filter equipment

Flowchart:

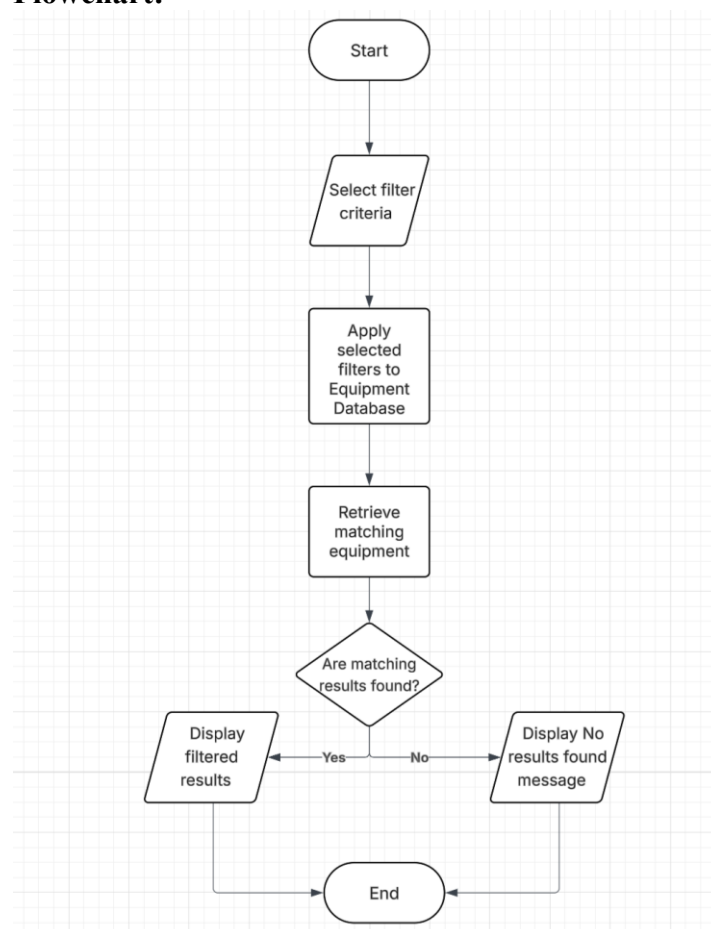


Figure 21: Flowchart for Filter Equipment

Pseudocode:

BEGIN FilterEquipment(selectedType, selectedPriceRange, selectedLocation)

**allEquipment = GET AllActiveEquipment()
INITIALIZE filteredResults as empty list**

FOR EACH item IN allEquipment DO

matchType = (selectedType == NULL OR item.type == selectedType)

matchPrice = (selectedPriceRange == NULL OR item.price WITHIN selectedPriceRange)

matchLocation = (selectedLocation == NULL OR item.location == selectedLocation)

IF (matchType AND matchPrice AND matchLocation) THEN

ADD item TO filteredResults

END IF

ENDFOR

IF (filteredResults IS NOT EMPTY) THEN

DISPLAY filteredResults

ELSE

DISPLAY "No results found"

END IF

END

Add Equipment

Flowchart:

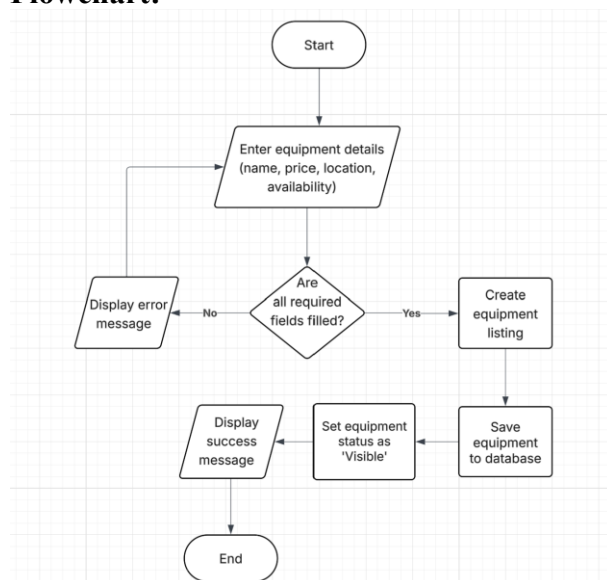


Figure 22 : Flowchart Add Equipment Listing

Pseudocode:

BEGIN AddEquipmentListing(ownerID)

isAdded = False

REPEAT

INPUT equipmentName, price, location, availability

**IF (equipmentName is empty OR
price is empty OR
location is empty OR
availability is empty) THEN**

DISPLAY "Error message"

```
ELSE
    CREATE new equipment listing
    SAVE equipment details to Database
    SET equipmentStatus = "Visible"
    DISPLAY "Equipment added successfully"
    isAdded = True
END IF
```

```
UNTIL isAdded = True
```

```
END
```

4.5 Interface Design

Site map :

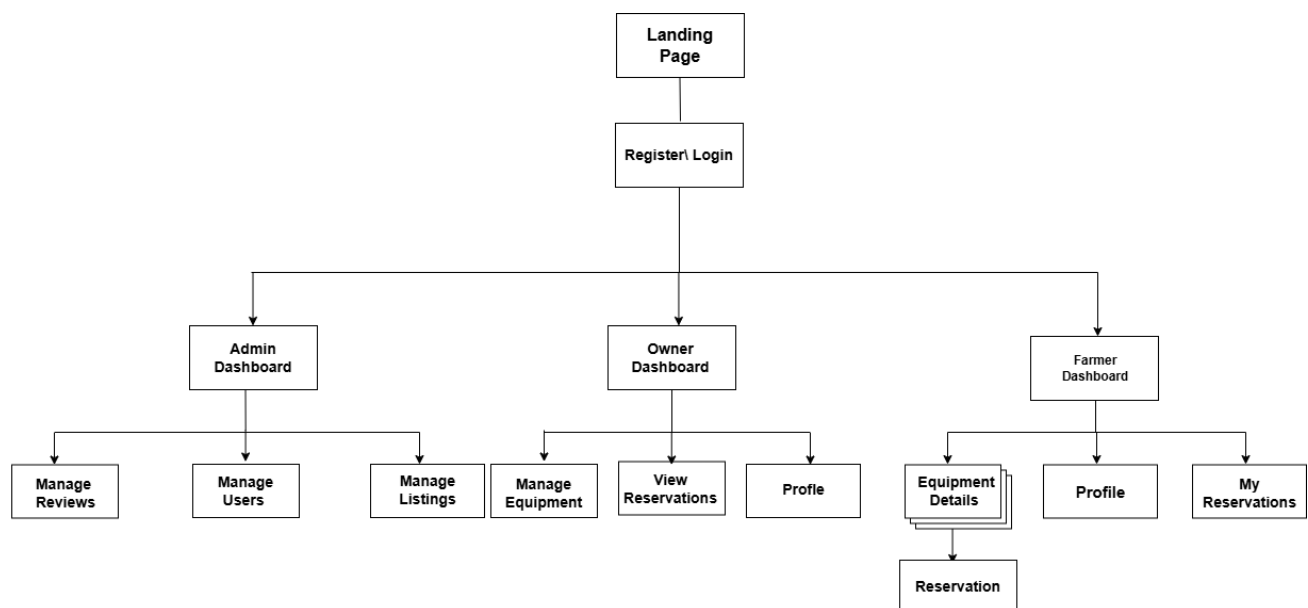


Figure 23 GreenRent System Site Map

Github link: <https://github.com/haya30/IT320-GreenRent-System>

Jira link : [jira](#)

The website : <https://haya30.github.io/IT320-GreenRent-System/>

5 Chapter 5: System Evaluation

This section brings together the findings from all testing phases conducted during the development of the GreenRent platform, including User Story Acceptance Testing, Integration Testing, and User Acceptance Testing. By analysing the results of these phases collectively, it becomes possible to evaluate how well the system fulfils its functional and non-functional requirements, and to identify areas that require further improvement.

5.1 User Story Acceptance Testing

User story acceptance testing determines whether a user story fulfills its purpose from the user's perspective. It answers the critical question: will the acceptance criteria be met upon execution of the user story?.

In this section, we outline how we conducted user story acceptance testing for the GreenRent platform to validate its overall functionality and efficiency. To ensure a thorough evaluation of each feature, we documented at least two test cases for every user story. We utilized a structured table to present the sprint number, the user story, the specific acceptance criteria, and the outcome of the test cases (pass or fail), along with any relevant comments.

Table 10: User Story Acceptance Testing Results

Sprint Number	User Story	Acceptance criteria	Test cases(s)	Pass?	Comments
2	As a new user, I want to register an account so that I can access rental services.	- The system displays a registration form with required fields (name, email, password, role). - The system validates that all required fields are completed.	Test case 1: (Success - Valid Registration)	Yes	-

		• If the email is already registered, the system shows an appropriate error message. • After successful registration, the user account is created. • The user is redirected to the login page (or dashboard). • The system stores the user information securely in the database.	1- User clicks on the Register option from the main page. 2- Fills all signup fields properly (Name, valid Email, Valid Phone Number ,Password, Role). 3- Clicks on the "Create Account" button. 4- Verifies that the account is created and a success message is displayed. Test case 2: (Failure - Duplicate Email) 1- User clicks on the Register option. 2- Fills the signup fields using an email that already exists in the system. 3- Clicks on the "Create Account" button. 4- Verifies that the system displays an error message stating that the email or phone number is already registered.		
2	As a registered user, I want to log in using my username and password so that I can securely access my account and personal information.	• The system displays a login form with username and password fields. • The system verifies the entered credentials against stored user data.	Test case 1: (Success - Valid Login) 1- User clicks on the Log In option.	Yes	

		• If the username and password are correct, the user is redirected to their dashboard. • If the credentials are incorrect, an appropriate error message is displayed. • The user session is created after successful login. • The user cannot access protected pages without logging in. • The login process should be completed within 5 seconds.	2- Enters a valid, registered email and the correct password. 3- Clicks the "Sign In" button. 4- Verifies that the system redirects to the appropriate dashboard. Test case 2: (Failure - Incorrect Password) 1- User clicks on the Log In option. 2- Enters a valid email but an incorrect password. 3- Clicks the "Sign In" button. 4- Verifies that access is denied and an "Incorrect email or password" error is shown.		
2	As a registered user, I want to log out securely so that I can protect my account from unauthorized access.	• The system provides a visible "Log out" option for logged-in users. • When the user clicks "Log out", the session is terminated immediately.	Test case 1: (Success - Standard Logout) 1- While logged in, user clicks the "Logout" button in the navigation bar. 2- Verifies that the session is terminated.	Yes	-

		• After logging out, the user is redirected to the login (or home) page. • The user cannot access protected pages after logout. • If the user tries to access a protected page after logout, the system redirects them to the login page. • The logout process completes successfully without errors.	3- Verifies that the user is redirected to the home/login page. Test case 2: (Failure - Unauthorized Access) 1- User logs out of the account successfully. 2- Attempts to access a protected dashboard URL directly via the browser address bar. 3- Verifies that the system denies access and redirects back to the login page.		
2	As a renter, I want to search equipment by name so that I can quickly find what I need.	• The system provides a search bar where the renter can enter equipment name keywords. • When a valid keyword is entered, matching equipment results are displayed. • If no equipment matches the keyword, the system displays a "No results found" message. • Search results are updated immediately after the renter submits the search query.	Test case 1: (Success - Valid Search) 1- Renter navigates to the Farmer Dashboard. 2- Enters the name of an existing equipment (e.g., "Tractor") in the search bar. 3- Verifies that the page updates and displays only the matching equipment cards. Test case 2:	Yes	-

			(Failure - No Results Found) 1- Renter navigates to the Farmer Dashboard. 2- Enters a random, non-existent keyword (e.g., "xyz123") in the search bar. 3- Verifies that the system displays a "No equipment matches your filters" message.		
2	As a renter, I want to filter equipment by type, price, and location so that I can narrow my choices.	• The system provides filter options for equipment type, price range, and location. • When the renter selects filter criteria, the displayed results update accordingly. • Multiple filters can be applied at the same time. • If no equipment matches the selected filters, a "No results found" message is displayed.	Test case 1: (Success - Apply Filters) 1- Renter navigates to the Farmer Dashboard. 2- Selects specific criteria from the dropdown menus (e.g., Type: "Tractor", Price: "Under SAR 200/day"). 3- Verifies that the dashboard updates automatically. 4- Verifies that only equipment matching all selected filters are displayed. Test case 2:	Yes	-

			(Failure - No Match Found) 1- Renter navigates to the Farmer Dashboard. 2- Selects a combination of filters that has no available equipment (e.g., an unlikely Type and Location). 3- Verifies that the equipment grid is empty. 4- Verifies that the system displays the "No equipment matches your filters" message.		
2	As a renter, I want to view detailed equipment information so that I can make a decision.	The system displays detailed equipment information, including name, description, price, location, and availability. The system displays the equipment condition and specifications clearly. The system displays images of the equipment. The equipment details page loads correctly when the renter selects an item.	Test case 1: (Success - View Details) 1- Renter browses the available equipment in the dashboard. 2- Clicks on a specific equipment card. 3- Verifies that the details page loads correctly. 4- Verifies that the image, price, location, condition, and specifications are displayed. Test case 2: (Failure - Invalid Item ID)	Yes	-

			1- Renter is logged into the system. 2- Attempts to navigate to an equipment details page with an invalid or deleted equipment ID in the URL (e.g., id=9999). 3- Verifies that the system handles the error gracefully. 4- Verifies that the system displays an "Equipment not found".		
2	As an equipment owner, I want to add my equipment details so that renters can view and rent it.	- The system provides a form where the owner can enter equipment name, description, price, location, and availability. - The system validates that all required fields are completed before submission. - After successful submission, the equipment is stored in the database. - The equipment becomes visible to renters in the listing page.	Test case 1: (successful equipment addition) 1- An equipment owner logs into the system 2- Navigates to the Manage Equipment page 3- The system displays a form with fields (equipment name, category, description, price, location, availability, condition, image) 4- The owner fills in all required fields with valid data	yes	-

			5- Clicks "Add Equipment" 6- The system inserts the equipment into the equipment table in the database 7- The equipment appears in the listings page and is visible to renters Test case 2: (missing required fields / invalid input) 1- An equipment owner logs into the system 2- Navigates to the Manage Equipment page 3- Leaves one or more required fields empty OR enters invalid data (e.g., price = 0) 4- Clicks "Add Equipment" 5- The system prevents submission and highlights invalid fields 6- An error message is displayed indicating that all required fields must be filled correctly		
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2	As an equipment owner, I want to edit my equipment details so that I can update information when needed.	• The system allows the owner to access and modify existing equipment details. • The system validates required fields before saving updates. • After successful update, the modified information is stored in the database. • The updated equipment details are immediately visible to renters.	Test case 1: (successful equipment update) 1- An equipment owner logs into the system 2- Navigates to the Manage Equipment page 3- Selects an existing equipment item from the edit dropdown/list 4- The system loads the equipment details into the form 5- The owner modifies one or more fields (e.g., price, location, description, availability) 6- Clicks "Save Changes" 7- The system updates the equipment record in the database 8- The updated equipment details are immediately reflected in the listings page Test case 2: (missing required fields during update)	yes	-
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			1- An equipment owner logs into the system 2- Navigates to the Manage Equipment page 3- Selects an existing equipment item 4- Deletes or leaves one or more required fields empty 5- Clicks "Save Changes" 6- The system prevents submission and highlights invalid fields 7- An error message is displayed indicating that required fields must be completed		
2	As an equipment owner, I want to delete my equipment so that it is no longer visible.	• The system allows the owner to select and delete a listed equipment item. • The system displays a confirmation message before deleting the equipment. • After confirmation, the equipment is permanently removed from the database. • The deleted equipment is no longer visible to renters on the platform	Test case 1: (Success - Confirm Deletion) 1- Owner locates an active equipment item in the dashboard. 2- Clicks the "Delete" button for that item. 3- Verifies that a confirmation message appears.	yes	-

			4- Clicks "Confirm" on the prompt. 5- Verifies that the equipment is permanently removed from the dashboard and is no longer visible to renters. Test case 2: (Failure - Cancel Deletion) 1- Owner locates an equipment item and clicks the "Delete" icon. 2- Verifies that the confirmation popup appears. 3- Clicks "Cancel" on the prompt. 4- Verifies that the deletion process is aborted. 5- Verifies that the equipment remains visible and active in the dashboard.		
2	As a renter, I want to reserve equipment for specific dates so that I can use it during my required period.	• The system allows the renter to select available start and end dates for the equipment. • The system prevents booking if the selected dates are already reserved. • After successful reservation, a confirmation message is displayed.	Test case 1: (Reserve on a valid date) 1- renter click on the desired equipment from the farmer's dashboard page	yes	-

		The reservation details are saved and visible to both the renter and the equipment owner.	2- renter navigates to the reservation page for an active piece of equipment. 3-select a valid start date (today or later) and an end date at least one day after the start date. 4-fill in the delivery location and payment details 5-click“Confirm Reservation.” 5- The system saves the reservation with status “pending”. 6-System displays the confirmation screen showing the booking reference number, rental period, and total amount. Test case 2: (date conflict prevention) 1-renter click on the desired equipment from the farmer’s dashboard page 2-renter attempts to book the same equipment for dates that overlap an existing confirmed reservation		
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			3-clicking “Confirm Reservation,” 4-system detects the conflict and displays an error message: “Sorry, this equipment is already booked for the selected dates. Please choose different dates.” 5-No duplicate reservation is inserted in the database		
2	As an equipment owner, I want to receive secure online payment so that I can ensure guaranteed transactions.	- The system allows equipment owners to enable online payment for their listings. - The owner receives a confirmation notification after successful payment. The transaction details are recorded in the system. - Payment is transferred to the owner’s registered account.	Test case 1: (online payment) 1-renter completes the reservation form 2-selects "Online Payment," enters valid card details (cardholder name, 16-digit card number, expiry MM/YY, CVV) 3-clicks "Confirm Reservation." 4-The system inserts a payment record into the payments table with payment_method = 'online payment' and payment_status = 'pending'. 5-The reservation confirmation screen is displayed and the transaction details (reservation ID, amount in SAR, payment method) are retrievable from the	yes	-

			database by the equipment owner Test case 2: (Bank transfer) 1- renter selects "Bank Transfer" as the payment method on the reservation page. 2-The system inserts a payment record with payment_method = 'bank transfer' and payment_status = 'pending'. 3- renter is shown bank transfer instructions on the confirmation screen. 4- The equipment owner can view the pending payment in their dashboard 5- the transaction record (reservation_id, amount, method, date) is stored in the payments table linked to the correct reservation		
2	As a renter, I want to leave a rating and review after renting so that I can share my experience.	• The system allows the renter to submit a rating (e.g., 1–5 stars) after completing a rental. • The renter can write and submit a written review. • Only renters who have completed a reservation can submit a rating and review. • The submitted rating and review are displayed on the equipment details page.	Test case 1: (successful review submission) 1-A renter whose reservation status is "completed" opens the My Reservations page 2- The reservation card shows a star-rating widget (1–5 stars) and a text area for a written review. 3- The renter selects 4 stars, types a written comment.	yes	-

			4- click" Submit Review" 5- The system inserts the review into the reviews table 6- the card now displays the submitted stars and review text instead of the rating form. Test case 2 : (trying to rate and review a not completed reservation) 1-A renter whose reservation status is "pending" or "confirmed" (not yet completed) 2- opens the My Reservations page 3- The review form (star widget and text area) is not shown for that reservation card , and there is a message saying " You can submit a rating after the rental is completed."		
2	As an Admin, I want to view and manage user accounts so that I can ensure proper use of the platform.	• The system allows the Admin to view a list of all registered users. • The system displays user details (name, email, role, account status). • The Admin can change the account status (active / inactive).	Test case 1: (view all registered users)	yes	

		- Any changes made by the Admin are saved in the database. - The updated account status is immediately reflected in the system.	1- An Admin logs into the system 2- Navigates to the User Accounts page 3- The system retrieves all users from the database 4- The system displays a list of all registered users 5- Each user record shows name, email, role, and account status Test case 2: (change user account status successfully) 1- An Admin logs into the system 2- Navigates to the User Accounts page 3- The Admin selects a user account 4- The Admin changes the account status (active → inactive or inactive → active) 5- The system updates the user status in the database 6- A success message is displayed		
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			7- The updated account status is immediately reflected in the user list Test case 3: (change user status with confirmation) 1- An Admin logs into the system 2- Navigates to the User Accounts page 3- The Admin clicks to change a user's account status 4- A confirmation message appears 5- The Admin confirms the action 6- The system updates the account status in the database Test case 4: (cancel status change action) 1- An Admin logs into the system 2- Navigates to the User Accounts page		
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			3- The Admin attempts to change a user's status 4- A confirmation message appears 5- The Admin clicks "Cancel" 6- The system does not update the account status 7- The user status remains unchanged Test case 5: (verify database update after status change) 1- An Admin logs into the system 2- Changes a user's account status 3- The system updates the users table in the database 4- The updated status is stored correctly 5- The change persists after page reload		
2	As an Admin, I want to suspend or deactivate user accounts when policy violations occur so that I can maintain platform integrity.	- The Admin can select a user account to suspend or deactivate. - The system requires confirmation before suspending the account.	Test case 1: (successful account suspension)	yes	

		• After confirmation, the user account status changes to “Suspended” or “Inactive”. • Suspended users cannot log in to the system. • The suspension action is recorded in the system	1- An Admin logs into the system 2- Navigates to the User Accounts page 3- The Admin selects a user account 4- The Admin chooses to suspend the account 5- A confirmation message appears 6- The Admin clicks "Confirm" 7- The system updates the user status to "Suspended" in the database 8- The updated status is displayed in the user list Test case 2: (successful account deactivation) 1- An Admin logs into the system 2- Navigates to the User Accounts page 3- The Admin selects a user account 4- The Admin chooses to deactivate the account		
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			5- A confirmation message appears 6- The Admin confirms the action 7- The system updates the user status to "Inactive" in the database 8- The updated status is immediately reflected in the system Test case 3: (cancel suspension or deactivation action) 1- An Admin logs into the system 2- Navigates to the User Accounts page 3- The Admin selects a user account 4- The Admin clicks to suspend or deactivate the account 5- A confirmation message appears 6- The Admin clicks "Cancel" 7- The system does not change the account status 8- The user status remains unchanged		
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			Test case 4: (restrict suspended user from logging in) 1- A user account is marked as "Suspended" 2- The user attempts to log in using valid credentials 3- The system checks the account status 4- The system denies access 5- An error message is displayed indicating the account is suspended Test case 5: (verify suspension action stored in database) 1- An Admin logs into the system 2- Suspends a user account 3- The system updates the users table in the database 4- The status is saved as "Suspended" or "Inactive"		
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			5- The change persists after page reload		
2	As an Admin, I want to review equipment listings to ensure they comply with platform policies.	• The system allows the Admin to view all equipment listings. • The Admin can access equipment details (owner, price, description, location, availability). • The system displays listing status (active / inactive) • The Admin can identify listings that violate platform policies.	Test case 1: (view all equipment listings) 1- An Admin logs into the system 2- Navigates to the Equipment Listings page 3- The system displays all equipment listings from all owners 4- Each row shows equipment details such as name, owner, price, location, condition, and availability status Test case 2: (search for specific equipment) 1- An Admin logs into the system 2- Navigates to the Equipment Listings page 3- Enters a keyword in the search field (e.g., equipment name or location) 4- The system filters the table results based	yes	

			on the entered keyword 5- Only matching equipment listings are displayed Test case 3: (restrict access for non-admin users) 1- An equipment owner logs into the system 2- Attempts to access the admin-listings.php page directly 3- The system checks the user role 4- Access is denied and the user is redirected to the login page 5- The admin listings page is not displayed		
2	As an Admin, I want to remove or deactivate equipment listings that receive repeated complaints or low ratings so that I can maintain service quality.	The Admin can view ratings and complaints related to each equipment listing.	Test case 1:		

		• The system highlights listings with consistently low ratings. • The Admin can deactivate or remove a listing. • The system requires confirmation before removal. • Removed listings are no longer visible to users. • The action is recorded in the system.	(view ratings and complaints for equipment) 1- An Admin logs into the system 2- Navigates to the Equipment Listings page 3- The system displays all equipment listings 4- The Admin views ratings and complaints related to each equipment listing Test case 2: (highlight listings with low ratings) 1- An Admin logs into the system 2- Navigates to the Equipment Listings page 3- The system identifies listings with consistently low ratings 4- The system highlights these listings visually 5- The Admin can easily distinguish problematic listings		
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			Test case 3: (successful equipment deactivation) 1- An Admin logs into the system 2- Navigates to the Equipment Listings page 3- The Admin selects an equipment listing 4- The Admin clicks "Deactivate" 5- A confirmation message appears 6- The Admin clicks "Confirm" 7- The system updates the listing status to "Inactive" in the database 8- The updated status is reflected in the listings page Test case 4: (successful equipment removal) 1- An Admin logs into the system 2- Navigates to the Equipment Listings page		
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			3- The Admin selects an equipment listing 4- The Admin clicks "Delete" 5- A confirmation message appears 6- The Admin confirms the action 7- The system deletes the equipment record from the database 8- The listing is no longer visible to users Test case 5: (cancel removal action) 1- An Admin logs into the system 2- Navigates to the Equipment Listings page 3- The Admin clicks "Delete" on a listing 4- A confirmation message appears 5- The Admin clicks "Cancel" 6- The system does not delete the listing		
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			7- The equipment remains visible in the list Test case 6: (verify action is recorded in system) 1- An Admin logs into the system 2- Performs a deactivate or delete action on a listing 3- The system updates the database accordingly 4- The change persists after page reload 5- The action is reflected in the system records		
2	As an Admin, I want to monitor reviews and ratings to identify misuse or policy violations.	• The system allows the Admin to view all submitted reviews and ratings. • The Admin can identify inappropriate or abusive content. • The Admin can take appropriate action if needed	Test case 1: (view all reviews and ratings) 1- An Admin logs into the system 2- Navigates to the Reviews & Ratings page 3- The system retrieves all submitted		

			reviews and ratings from the database 4- The system displays all reviews with their ratings 5- Each review shows relevant details (user, equipment, rating, comment) Test case 2: (identify inappropriate or abusive content) 1- An Admin logs into the system 2- Navigates to the Reviews & Ratings page 3- The Admin browses through submitted reviews 4- The Admin identifies a review containing inappropriate or abusive content 5- The system allows the Admin to recognize problematic reviews clearly Test case 3: (take action on inappropriate reviews)		
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			1- An Admin logs into the system 2- Navigates to the Reviews & Ratings page 3- The Admin selects a review identified as inappropriate 4- The Admin takes an action (e.g., remove or flag the review) 5- The system updates the database accordingly 6- The updated result is reflected in the system Test case 4: (no action needed for valid reviews) 1- An Admin logs into the system 2- Navigates to the Reviews & Ratings page 3- The Admin reviews submitted feedback 4- The Admin determines the review is appropriate 5- No action is taken		
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			6- The review remains visible in the system Test case 5: (verify review data persistence) 1- An Admin logs into the system 2- Views reviews and ratings 3- The system retrieves data from the database 4- The displayed reviews match the stored data 5- Data remains consistent after page reload		
2	As a renter, I want search results to load within 3 seconds so that I can quickly find equipment	• Search results load within 3 seconds under normal load. • System performance is tested before deployment.	Test case 1: (under normal load) 1-User click the search bar in the farmer dashboard 2-User enters a keyword (e.g., "tractor") 3- User clicks search 4-Measure loading time(1.2 sec) Test case 2 : (applies multiple simultaneous filters on the dashboard (e.g., equipment type, location, and availability).) 1-User click the search bar in the farmer dashboard	yes	loaded in 1.2 sec

			2-User enters a keyword (e.g., "tractor") and apply multiple filters 3- User clicks search 4- The system executes the filtered SQL query against the equipment and reservations tables. 5-Measure loading time(1.2 sec)		
2	As a user, I want the platform to be available at least 99% of the time so that I can access rental services whenever I need them	• The system uptime must be at least 99% per month. • Scheduled maintenance must be announced in advance. • In case of downtime, users receive a notification message. • The system automatically logs downtime incidents.	Test case 1 : (single user availability) 1- Open the browser 2- enter in the browser url the "GreenRent" url 3-check if the page is available Test case 2 : (concurrent user access) 1- have 6 users to enter the "GreenRent" url at the same time 2- check if the "landing page " is available to the users	yes	-

5.2 Integration Testing

Integration testing was conducted to verify that the main components of the GreenRent system work correctly after being integrated into one complete web-based platform. This testing focused on the

interaction between authentication, role-based dashboards, equipment details, reservation processing, profile pages, admin user management, reviews and ratings management, and the database. The main purpose was to ensure that users can move between connected pages successfully, that each role is redirected to the correct dashboard, and that data is stored, retrieved, and updated correctly across the system.

5.2.1 Integration Hierarchy

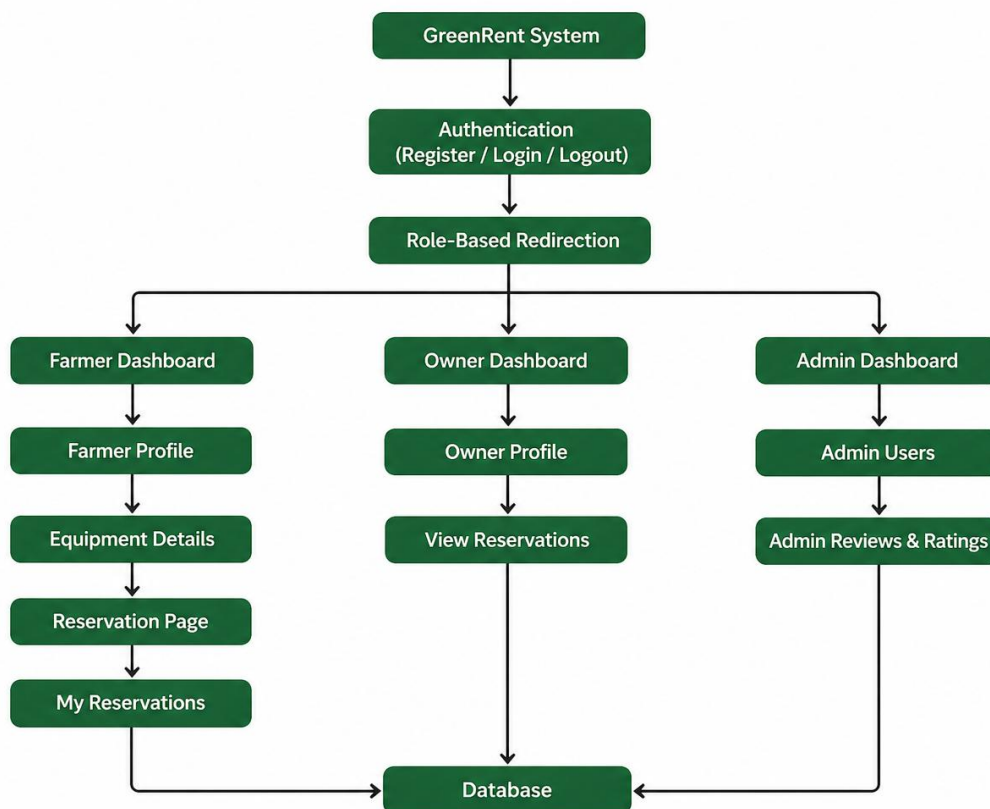


Figure 24 Integration hierarchy

The integration hierarchy shows the execution flow between the main GreenRent components. The system starts with authentication, where users register, log in, or log out. After successful login, the system redirects the user based on their role. Farmers are directed to the Farmer Dashboard, where they can view equipment details, make reservations, and access their profile and reservation history. Equipment owners are directed to the Owner Dashboard, where they can view their profile and reservations related to their listed equipment. Admins are directed to the Admin Dashboard, where they can manage users and monitor reviews and ratings. The database supports all these flows by storing and retrieving users, equipment, reservations, reviews, and payment data.

5.2.2 Integration Testing Plan

Scenario 1: Farmer Reservation Flow

Table 11: Farmer Reservation Flow Test Cases

System components	New component	Test case	Pass?	Comments
Logging in the system	Farmer Dashboard	1. A farmer logged in successfully using valid credentials. 2. The system redirected the farmer to the Farmer Dashboard based on the user role.	Yes	The role-based redirection worked correctly.
Logging in the system, Farmer Dashboard	Equipment Details	1. The farmer selected an equipment item from the dashboard. 2. The system opened the Equipment Details page. 3. The selected equipment information was displayed correctly.	Yes	Equipment data was retrieved correctly from the database.
Farmer Dashboard, Equipment Details	Reservation Page	1. The farmer clicked the Reserve option. 2. The system opened the Reservation Page. 3. The farmer entered reservation dates and payment information.	Yes	The reservation form was displayed and completed successfully.
Equipment Details, Reservation Page, Database	My Reservations	1. The farmer submitted the reservation. 2. The reservation details were saved in the database. 3. The farmer opened My Reservations and the new reservation appeared.	Yes	Reservation data was stored and retrieved correctly.

Scenario 2: Admin Management Flow

Table 12: Admin Dashboard and User Management Test Cases

System components	New component	Test case	Pass?	Comments
Logging in the system	Admin Dashboard	1. An admin logged in successfully using valid credentials. 2. The system redirected the admin to the Admin Dashboard based on the user role.	Yes	The admin role was detected correctly.
Logging in the system, Admin Dashboard	Admin Users	1. The admin selected the Admin Users page. 2. The system displayed all registered users with their names, emails, roles, and account statuses.	Yes	User data was retrieved correctly from the database.
Admin Dashboard, Admin Users	Account Status Update	1. The admin selected a user account. 2. The admin changed the account status to suspended or active. 3. The updated status was saved in the database.	Yes	Account status was updated and reflected in the system.
Admin Dashboard	Admin Reviews & Ratings	1. The admin opened the Reviews & Ratings page. 2. The system displayed all submitted reviews and ratings. 3. The admin reviewed flagged or low-rated feedback.	Yes	Reviews and ratings were retrieved correctly from the database.

5.3 User Acceptance Testing

The User Acceptance Testing (UAT) was conducted to understand how users interact with the GreenRent system and to evaluate their overall experience while using it. A total of 10 participants were involved in this testing process.

During the test, participants were asked to perform several tasks, such as navigating between different pages, completing a reservation, and exploring the system features. After finishing these tasks, they were asked to provide their feedback through a questionnaire that focused on usability, clarity, navigation, and overall satisfaction.

5.3.1 Demographics of Participants

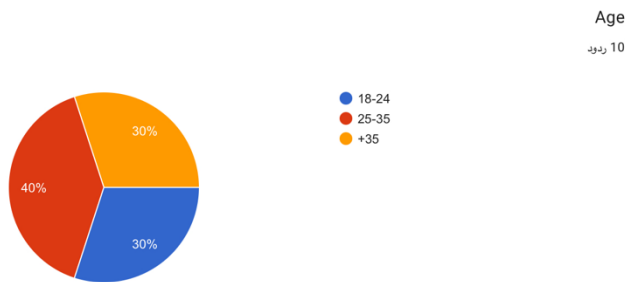


Figure 25 : Demographics of Participants - Age

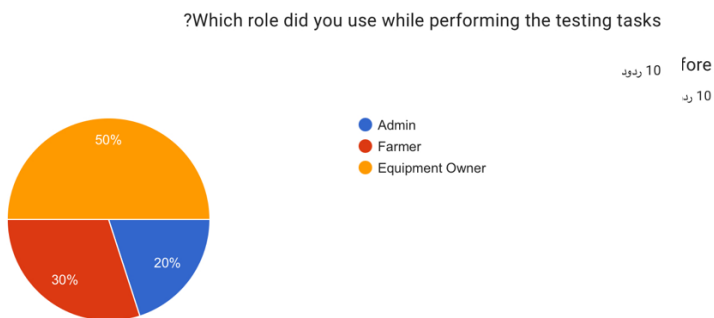


Figure 26 : Demographics of Participants – which role did you use while performing the testing tasks ?

In this section, we show the demographic information of the participants who took part in the testing. This includes their age groups, the role they used during the test, and whether they have used online booking systems before.

The results show that the participants were from different age groups, which helped us get more varied feedback. Also, participants used different roles such as Admin, Farmer, and Equipment Owner, which helped in testing the system from different perspectives.

In addition, some participants had previous experience with online booking systems, while others did not. This helped us understand whether the system is easy to use for both experienced users and new users.

5.3.2 Questionnaire Results

In this part, we show the participants' responses to the questionnaire based on their experience with the system. The questions were designed to evaluate usability, navigation, performance, and overall satisfaction using a scale from 1 to 5, where 5 indicates the highest satisfaction level.

Questionnaire Questions:

1. The system is easy to use.
2. The interface design is clear and well-organized.
3. I was able to complete tasks without confusion.
4. Navigation between pages is smooth and easy.
5. The system responds quickly to user actions.
6. The process of reserving equipment is simple and straightforward.
7. The information provided in the system is clear and understandable.
8. The system features are easy to understand.
9. Overall, I am satisfied with the system.

نسخ الرسم البياني

The system is easy to use

10 ردود

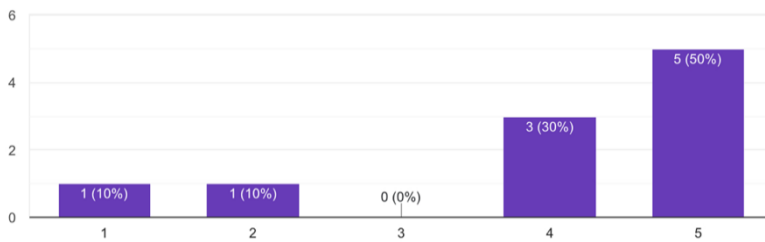


Figure 27 : Questionnaire Question 5.3.2- The system is easy to use

نسخ الرسم البياني

The interface design is clear and well-organized

10 ردود

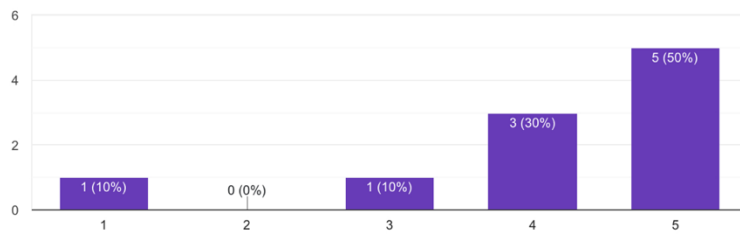


Figure 28 : Questionnaire Question 5.3.2- The interface design is clear and well-organized

نسخ الرسم البياني

I was able to complete tasks without confusion

10 ردود

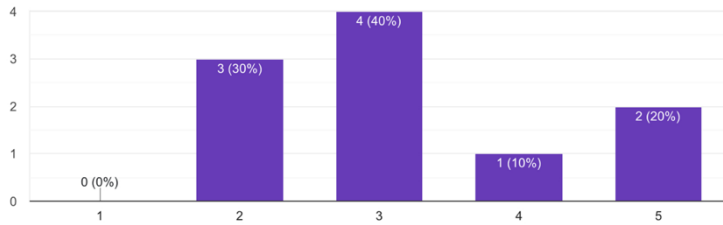


Figure 29 : Questionnaire Question 5.3.2- i was able to complete tasks without confusion

نسخ الرسم البياني

.Navigation between pages is smooth and easy

10 ردود

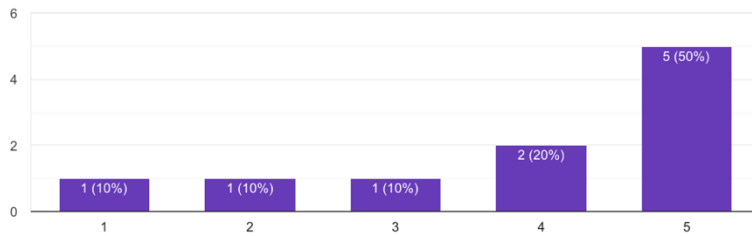


Figure 30 : Questionnaire Question 5.3.2- Navigation between pages is smooth and easy

نسخ الرسم البياني

The process of reserving equipment is simple and straightforward

10 ردود

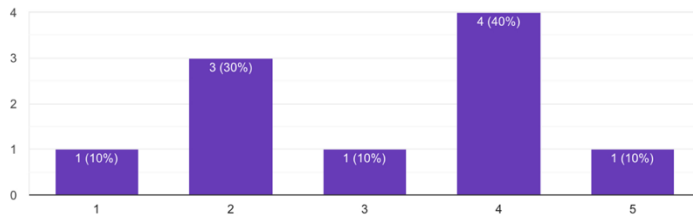


Figure 31 : Questionnaire Question 5.3.2- the process of reserving equipment is simple and straightforward

نسخ الرسم البياني

The system responds quickly to user actions

10 ردود

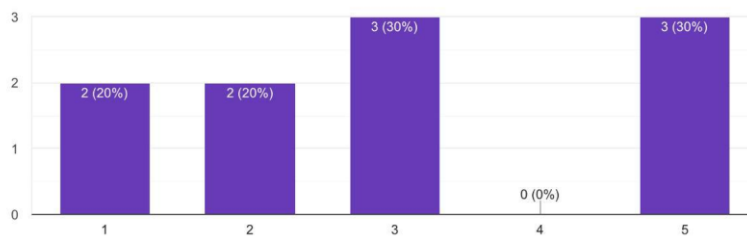


Figure 32 : Questionnaire Question 5.3.2- the system action's respond

نسخ الرسم البياني

.The information provided in the system is clear and understandable

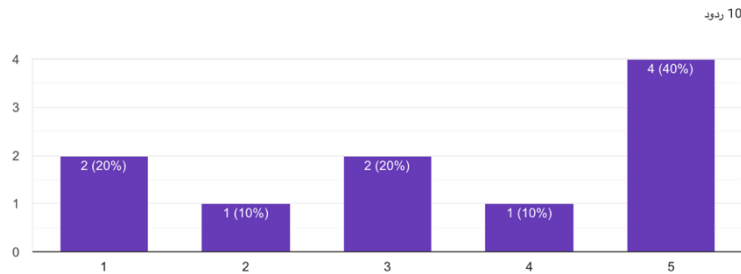


Figure 34 : Questionnaire Question 5.3.2- The information is understandable and clear

نسخ الرسم البياني

The system features are easy to understand

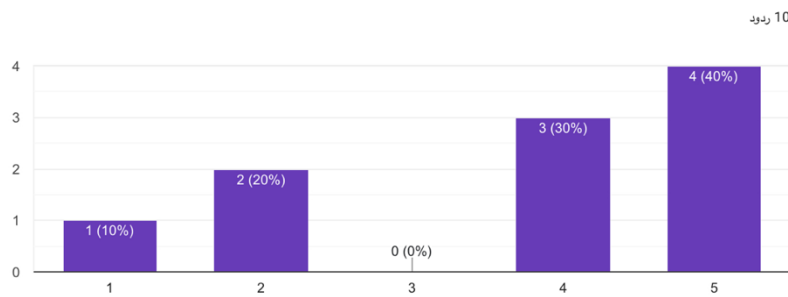


Figure 33 : Questionnaire Question 5.3.2- the system features are easy to understand

نسخ الرسم البياني

Overall, I am satisfied with the system

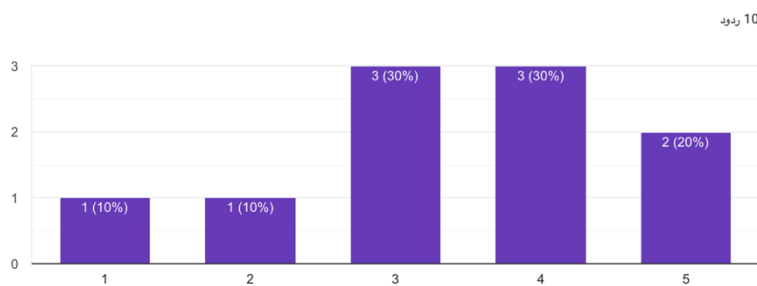


Figure 35 : Questionnaire Question 5.3.2- overall, i am satisfied with the system

Results Explanation

The results obtained from the questionnaire show that participants had a generally positive experience while using the system, although some aspects still need improvement.

Ease of Use:

Based on the results, 50% of the participants strongly agreed and 30% agreed that the system is easy to use. However, a small percentage (20%) gave lower ratings (1 and 2), which indicates that while most users found the system simple, a few users experienced some difficulty.

Interface Design:

The majority of participants rated the interface design positively, with 50% strongly agreeing and 30% agreeing that it is clear and well-organized. Only a small number of participants gave lower ratings, which suggests that the design is generally effective but could still be improved for better clarity.

Task Completion:

The responses for completing tasks without confusion were more varied. Around 40% of participants selected a neutral rating (3), while others were distributed between higher and lower ratings. This indicates that some users were able to complete tasks easily, while others experienced confusion, suggesting a need for improving certain functionalities.

Navigation:

Navigation between pages was evaluated positively, with 50% of participants strongly agreeing that it is smooth and easy. However, some lower ratings were also observed, indicating that navigation is generally good but may not be fully intuitive for all users.

System Responsiveness:

The responses for system responsiveness were mixed. While 30% of participants gave the highest rating (5) and another 30% selected (3), there were also users who gave lower ratings (1 and 2). This suggests that the system performance is inconsistent and may require optimization.

Reservation Process:

The process of reserving equipment received moderately positive feedback. The highest percentage (40%) rated it as 4, while the rest of the responses were spread across other ratings. This indicates that the process is understandable, but not fully smooth for all users.

Information Clarity:

The clarity of the information provided in the system showed mixed results. Although 40% of participants gave the highest rating (5), there were also noticeable lower ratings. This suggests that while some users found the information clear, others may have faced difficulty understanding it.

System Features:

Most participants found the system features easy to understand, with 40% selecting rating 5 and 30% selecting rating 4. However, a few lower ratings indicate that some features may need further clarification.

Overall Satisfaction:

Overall satisfaction results were moderate rather than very high. Most responses were concentrated at ratings 3 and 4, with fewer participants selecting 5. This indicates that users are generally satisfied with the system, but there is still room for improvement to achieve higher satisfaction levels.

5.4 Discussion

This section provides a detailed analysis of the results collected during the testing phase.

It also includes reflections on how the system was developed, focusing on the design choices, implementation, and the overall process.

5.4.1 Interpretation of results

The analysis of the questionnaire results shows that the system successfully achieved its primary functional requirements, particularly in terms of ease of use and navigation. Approximately **70% of participants** reported that the system was easy to use, indicating that the interface design effectively supports different types of users.

Additionally, the high ratings for interface clarity suggest that the design decisions, such as structured layout and clear feature organization, contributed significantly to improving usability.

However, the results related to task completion and system responsiveness were comparatively lower. This indicates that while the system is generally usable, it does not fully satisfy the requirement of speed, as some users experienced delays or required additional effort to complete tasks.

5.4.2 Reflection on the Process

During the development process, the work was divided among the team members, which helped in organizing the tasks and completing the project more efficiently. Each member was responsible for specific parts of the system, which made it easier to manage the workload.

However, there were some challenges during the process, especially in coordinating between different parts of the system. In some cases, integrating different features required additional effort to ensure everything works correctly together.

Despite these challenges, the process helped improve teamwork and communication between the team members. It also highlighted the importance of planning and testing during development to avoid issues later.

5.4.3 Reflection on Design and Implementation

Based on the testing results, the system design was generally clear and easy to understand. Most users were able to interact with the interface without major issues, which indicates that the design was successful in supporting usability.

The layout of the system helped users navigate between different sections smoothly, and this reflects that the structure of the system was well organized. In addition, the main features of the system were understandable for most users.

However, some users experienced difficulty in completing certain tasks, which suggests that a few parts of the system may not be fully intuitive. This shows that while the design is effective overall, there is still room for improvement.

From an implementation perspective, the system was able to achieve its main functionalities, such as navigation and reservation. However, the feedback related to system responsiveness indicates that performance can be improved to provide a smoother user experience.

5.4.4 Reflection on Results (UAT & NFR)

From a requirements perspective, the system successfully meets the **ease of use requirement**, as most users were able to interact with the system without major difficulties. However, the **speed requirement was only partially achieved**, due to moderate feedback on responsiveness.

This suggests that while the system design is effective, performance optimization is necessary to fully satisfy non-functional requirements related to system efficiency.

5.4.5 Overall Discussion

Overall, the GreenRent system can be considered functionally successful, as it meets the main user needs and supports core operations such as searching, filtering, and reservation. The system also demonstrates strong usability, supported by positive user feedback.

However, the evaluation results highlight important areas for improvement, particularly in system responsiveness and task clarity. Addressing these issues would significantly enhance user satisfaction and ensure that both functional and non-functional requirements are largely achieved.

These findings confirm that the system design decisions were effective in supporting usability and meeting user expectations.

6 Chapter 6: Conclusion and Future Work

6.1 Conclusion

The GreenRent system can be considered largely successful, as it was able to achieve its main objectives of providing a user-friendly and accessible platform for equipment rental. The evaluation results indicate that most users were able to interact with the system easily, navigate between different pages smoothly, and understand the main features without significant difficulties. This reflects that the design decisions, particularly in terms of layout and feature organization, effectively supported usability and user interaction.

However, the results also revealed some areas that require further improvement. In particular, system responsiveness and task completion without confusion did not receive equally strong feedback compared to other aspects. This suggests that while the system meets its core functional requirements,

it does not fully satisfy certain non-functional requirements, especially those related to speed and clarity. Therefore, although the overall user experience is positive, enhancing performance and simplifying some processes would be essential to further improve user satisfaction and system effectiveness.

6.2 Future Work

For future work, several enhancements can be considered to improve both system performance and usability. One of the key priorities is optimizing system responsiveness to better meet the speed requirement and provide a smoother user experience. In addition, simplifying certain workflows and making system features more intuitive would help reduce user confusion during task completion. Providing clearer instructions or visual guidance within the interface could also support users, especially those who are less familiar with the system.

Furthermore, the system can be expanded by introducing additional features such as real-time availability updates, improved feedback mechanisms, and more advanced filtering or recommendation options. Enhancing the system's scalability and performance under higher usage conditions would also be important for future deployment. Overall, these improvements would strengthen the system and ensure the system achieves higher efficiency, usability, and overall user satisfaction.

7 References

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7.1 Appendix A: Interview

This Appendix will represent all the interview-related supplements, including the interview questions and transcriptions.

• Interview Questions:

1. Do you currently rent out your agricultural equipment? If yes, how do you manage the process?
2. What challenges or risks do you foresee if you were to offer your agricultural equipment for rent to other farmers?
3. What factors would influence your decision to rent out your equipment through an online platform?
4. What information should be displayed about the equipment?
5. Is it important for you to have a rating system for renters? Why?
6. What payment method would you prefer (bank transfer, online payment, cash)?
7. What concerns do you have about using an online rental platform?

- Interview's Transcriptions:

The interview transcriptions provide a written record of the conversations conducted with participants during the requirements elicitation phase. Each transcription includes the interview details, questions, and participants' responses. These records help ensure accuracy in understanding user needs and support the identification of functional and non-functional requirements for the GreenRent system.

Table 13 : Interview 1

Interview (1)	
Field	Details
Interviewer	Haya
Interviewee	Ali
Date	19 February 2026
Q1: Do you currently rent out your agricultural equipment? If yes, how do you manage the process?	No, I do not currently rent out my equipment. I only use it for my own farm operations and have never offered it for rent to others.
Q2: What challenges do you think you might face if you decided to rent out your agricultural equipment?	I think the main challenges would be the risk of equipment damage and making sure it is returned on time. I would also worry about dealing with renters who may not handle the equipment properly or delay payments. Another challenge could be managing bookings and ensuring the equipment is available when I need it for my own farm work.
Q3: What factors would influence your decision to rent out your equipment through an online platform?	A secure payment system, verified renters, clear rental agreements, and protection against damage would encourage me to consider renting.
Q4: What information should be displayed about the equipment?	Equipment type, specifications, rental price, availability dates, condition, maintenance history, and location should be clearly displayed.
Q5: Do you think a rating system based on renter's feedback is important? Why?	Yes, I think it is important. If renters leave feedback about their experience, it helps build trust. Good ratings would make other customers feel more comfortable renting from me. It also helps me understand if there is anything I need to improve.
Q6: What payment method would you prefer (bank transfer, online payment, cash)?	I prefer online payment through the platform or bank transfer to ensure secure and documented transactions.
Q7: What concerns do you have about using an online rental platform?	I am concerned about fraud, unclear terms and conditions, and the technical complexity of the platform.

Table 14 : Interview 2

Interview (2)	
Field	Details
Interviewer	Sara
Interviewee	Mohammed
Date	20 February 2026
Q1: Do you currently rent out your agricultural equipment? If yes, how do you manage the process?	No
Q2: What challenges or risks do you foresee if you were to offer your agricultural equipment for rent to other farmers?	I would need additional labor to manage the equipment. There would also be maintenance costs. Dealing with renters could be difficult, as we might agree on a specific time, but they may fail to commit, which could result in losing other rental opportunities. It might also not be profitable if maintenance costs become higher than the rental income.
Q3: What factors would influence your decision to rent out your equipment through an online platform?	Trust in the platform is very important. I need to be sure that I will receive my payment. I am also concerned about not knowing the person who books the equipment and whether they are able to pay.
Q4: What information should be displayed about the equipment?	It should clearly state whether the equipment comes with an operator. The platform should also provide detailed information about the machine, including its condition and performance capability.
Q5: Do you think a rating system based on renter's feedback is important? Why?	Yes, it is important. It helps me assess the renter's credibility and determine whether the equipment performs as described.
Q6: What payment method would you prefer (bank transfer, online payment, cash)?	I prefer online payment after the equipment has been used.
Q7: What concerns do you have about using an online rental platform?	I am concerned about the quality of the equipment and the fairness of pricing.

Table 15 : Interview 3

Interview (3)	
Field	Details
Interviewer	Deema
Interviewee	Abdulaziz
Date	21 February 2026
Q1: Do you currently rent out your agricultural equipment? If yes, how do you manage the process?	Yes, I sometimes rent them to neighboring farmers. I manage it personally and informally through verbal agreements on price and duration, without official contracts.
Q2: What challenges or risks do you foresee if you were to offer your agricultural equipment for rent to other farmers?	The biggest challenge is misuse leading to costly breakdowns, and the renter returning the equipment late, which disrupts the work on my own farm
Q3: What factors would influence your decision to rent out your equipment through an online platform?	Having clear guarantees to protect my rights in case of equipment damage, an easy-to-use platform, and the availability of electronic contracts to protect both parties.
Q4: What information should be displayed about the equipment?	Equipment type, model, maintenance condition, daily rental price, location, and clearly stating whether it is rented with or without an operator.
Q5: Do you think a rating system based on renter's feedback is important? Why?	Yes, it is very important. Positive reviews from renters build trust and bring me more customers. Their feedback also helps me improve the service and fix any shortcomings.
Q6: What payment method would you prefer (bank transfer, online payment, cash)?	Prepaid online payment through the platform or bank transfer. This secures my rights, documents the financial transaction, and avoids the awkwardness of asking for cash.
Q7: What concerns do you have about using an online rental platform?	I am concerned about the difficulty of contacting technical support if a problem occurs, and that the platform fees might be too high, reducing my profit.

Table 16 : Interview 4

Interview (4)	
Field	Details
Interviewer	Nourah
Interviewee	Ahmad
Date	22 February 2026
Q1: Do you currently rent out your agricultural equipment? If yes, how do you manage the process?	No, I do not currently rent out my equipment. I mainly use it for my own farming activities, but I am interested in renting it in the future if there is a secure and organized system.
Q2: What challenges or risks do you foresee if you were to offer your agricultural equipment for rent to other farmers?	I am concerned about equipment damage, late returns, and whether renters will use the machinery properly. Another challenge is ensuring that the equipment is available when I personally need it.
Q3: What factors would influence your decision to rent out your equipment through an online platform?	Having a trusted platform with verified users, clear rental policies, insurance or damage protection, and guaranteed payment would strongly influence my decision.
Q4: What information should be displayed about the equipment?	The platform should display the equipment type, model, condition, rental price, availability schedule, location, and whether the equipment requires an operator.
Q5: Do you think a rating system based on renter's feedback is important? Why?	Yes, it is very important. Ratings help build trust between users and allow equipment owners to decide whether a renter is reliable or not.
Q6: What payment method would you prefer (bank transfer, online payment, cash)?	I prefer online payment through the platform because it is secure, documented, and reduces the risk of payment issues.
Q7: What concerns do you have about using an online rental platform?	I am concerned about technical issues, customer support response time, and whether the platform fees would reduce my overall profit.

Table 17 : Interview 5

Interview (5)	
Field	Details
Interviewer	Dalal
Interviewee	Saleh
Date	20 February 2026
Q1: Do you currently rent out your agricultural equipment? If yes, how do you manage the process?	Yes, I sometimes rent out my equipment, but only to farmers I know personally. We usually agree on the price and duration verbally. There is no written contract, and the process depends mostly on trust between us.
Q2: What challenges or risks do you foresee if you were to offer your agricultural equipment for rent to other farmers?	The main risks I see are equipment damage, late returns, or not receiving full payment. Agricultural machines are expensive, and any serious damage can cost a lot to repair
Q3: What factors would influence your decision to rent out your equipment through an online platform?	I would consider using an online platform if it protects my rights. I would need secure payment, verified users, and a clear rental agreement. The platform should also be simple and easy to use.
Q4: What information should be displayed about the equipment?	It should show the equipment type, condition, price, and availability clearly so there is no misunderstanding between the owner and the renter.
Q5: Do you think a rating system based on renter's feedback is important? Why?	Yes, a rating system is important. It helps me know if the renter is responsible and trustworthy before I agree to rent my equipment.
Q6: What payment method would you prefer (bank transfer, online payment, cash)?	I prefer bank transfer or secure online payment through the platform. I prefer having a clear payment record.

Q7: What concerns do you have about using an online rental platform?

My main concerns are fraud, equipment misuse, and difficulty solving disputes. I would only use the platform if it is secure and reliable.

What improvements or additional features would you suggest for an agricultural equipment rental website

10 ردود

direct chat between renter and owner

add availability calendar

customer support hotline

offer seasonal discounts

secure online payment system

Provide insurance options for equipment damage

.delivery service for large machines

clear rental agreements

Figure 36 : **What improvements or additional features would you suggest?** Gathers ideas for future platform improvements.

Graph 10

Questionnaire 2 : Equipment owner

1. Have you ever rented out your agricultural equipment to others?
2. What is the biggest challenge or main concern that might prevent you from renting out your agricultural equipment?
3. As an equipment owner, what is the most important technical feature you look for in the platform?
4. Which of the following options would make you feel safer when listing your equipment on the platform?
5. What is your preferred method of receiving rental profits?
6. How often do you expect your equipment to be available for rent through the platform?
7. Open question, What suggestions or additional services would you like to see on the platform to facilitate the process of renting out your equipment?

Figure 37 : Summary of Suggested Additional Services to Facilitate the Rental Process

Graph 17

Ability to offer discounts for long rental periods
I need a direct way to chat with the renter before accepting
.Automate SMS reminders for renters before the end of the renter period
Include location tracking for the equipment
Provide a dedicated customer service number for emergencies
Detailed maintenance logs tracking within the platform
Option to require the equipment to be rented with my own driver
Clear guidelines on who is responsible for repairing the equipment if it breaks down during the rental period
A feature to block certain users if they cause trouble or return equipment late

7.2 Appendix C : Jira's Backlog

The screenshot displays the Jira Product Backlog for the 'GreenRent' project. The interface includes a left sidebar with navigation options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', 'Recent', 'More spaces', 'Recommended', 'Collect requests', 'Filters', 'Dashboards', 'Confluence', 'Goals', and 'Teams'. The main area shows the 'GreenRent' project with a search bar and a 'Filter' button. Below this, a list of 19 work items is displayed, each with a title, priority, and status. The items are organized into a backlog view with a 'Create sprint' button at the top right.

ID	Title	Priority	Status	Assignee
SCRUM-5	As a new user, I want to register an account so that I can access rental services.	High	TO DO	Assignee
SCRUM-6	As a registered user, I want to log in using my username and password	High	TO DO	Assignee
SCRUM-7	As a registered user, I want to log out securely so that I can protect my account from unauthorized access.	High	TO DO	Assignee
SCRUM-8	As an equipment owner, I want to add my equipment details so that renters can view and rent it.	High	TO DO	Assignee
SCRUM-9	As an equipment owner, I want to edit my equipment details so that I can update information when needed.	High	TO DO	Assignee
SCRUM-10	As an equipment owner, I want to delete my equipment so that it is no longer visible.	High	TO DO	Assignee
SCRUM-11	As a renter, I want to search equipment by name so that I can quickly find what I need.	High	TO DO	Assignee
SCRUM-12	As a renter, I want to view detailed equipment information so that I can make a decision.	High	TO DO	Assignee
SCRUM-13	As a renter, I want to filter equipment by type, price, and location so that I can narrow my choices.	High	TO DO	Assignee
SCRUM-14	As a renter, I want to reserve equipment for specific dates so that I can use it during my required period.	High	TO DO	Assignee
SCRUM-15	As an equipment owner, I want to receive secure online payment so that I can ensure guaranteed transactions.	High	TO DO	Assignee
SCRUM-16	As a renter, I want to leave a rating and review after renting so that I can share my experience.	High	TO DO	Assignee
SCRUM-19	As an Admin, I want to view and manage user accounts so that I can ensure proper use of the platform.	High	TO DO	Assignee
SCRUM-20	As an Admin, I want to suspend or deactivate user accounts when policy violations occur so that I can maintain platform int...	High	TO DO	Assignee
SCRUM-21	As an Admin, I want to review equipment listings to ensure they comply with platform policies.	High	TO DO	Assignee
SCRUM-22	As an Admin, I want to remove or deactivate equipment listings that receive repeated complaints or low ratings so that I can...	High	TO DO	Assignee
SCRUM-23	As an Admin, I want to monitor reviews and ratings to identify misuse or policy violations.	High	TO DO	Assignee
SCRUM-17	As a renter, I want search results to load within 3 seconds so that I can quickly find equipment.	High	TO DO	Assignee
SCRUM-18	As a user, I want the platform to be available at least 99% of the time so that I can access rental services whenever I need t...	High	TO DO	Assignee

Figure 38 : Jira Product Backlog-Sprint 0

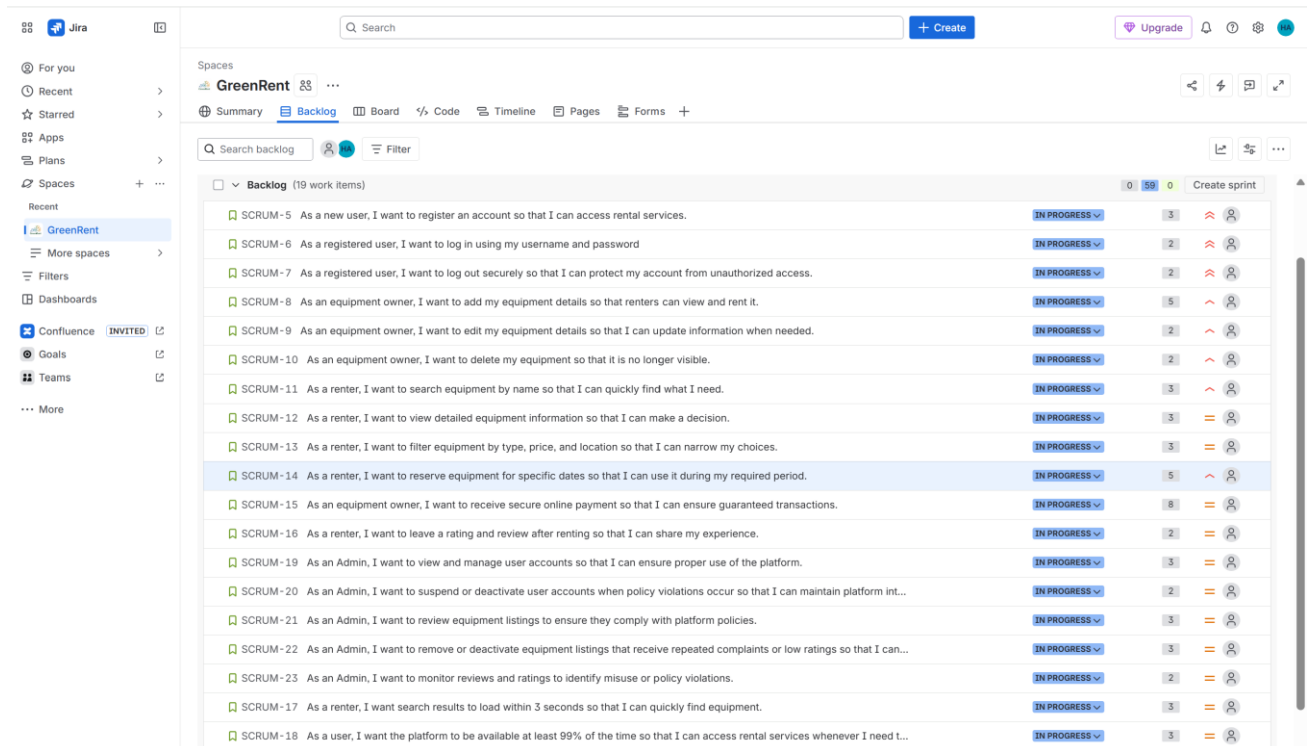


Figure 39 : Jira Product Backlog-Sprint 1

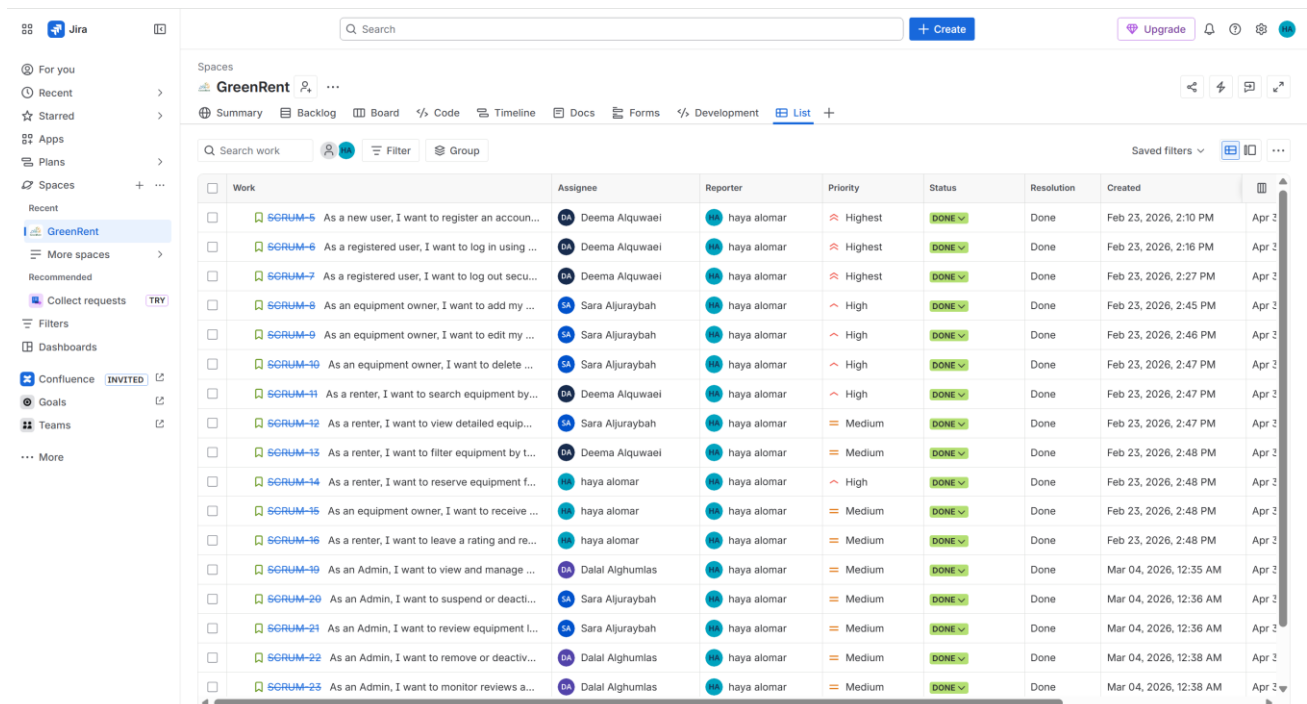


Figure 40 : Jira Product Backlog-Sprint 2

